Substances that are reabsorbed are **glucose, water**, and some other useful minerals. This occurs because our body cannot afford to lose there substances.

The process of reabsorption occurs in renal tubules.
These are:

- ✓ **First coiled (proximal) tubule**
- ✓ **Loop of Henle**
- ✓ **Second coiled (Distal) tubule**

i) Reabsorption in first coiled (proximal) tubule

Most of the reabsorption of takes place here. This involves.

- ✓ 67% of sodium ions by active transport
- ✓ 80% of water by osmosis
- ✓ 100% of the gulucose by active transport

This section has increased surface area because it has long and coiled portions.

ii) Loop of Henle

This is where the urine becomes concentrated and additional amount of water is reabsorbed.
In addition, loop of Henle is a place where the PH of the blood is maintanned. Because of this process, the PH of the blood becomes neutral (7.3 – 7.4).

iii) Second coiled (distal) tubule

This is where water gets balanced (osmoregulation).

- If the body is in a shortage or need of water, the pituitary gland is stimulated and release a hormone called **anti duretic hormone (ADH).**

ADH stimulates the reabsorption of more water in to the blood.

In addition, reabsorption of salt needed in the body takes place in distal tubule.

ADH (Anti diuretic hormone) is one of the hormones produced by the pituitary gland. It reduces the production of urine in the kidneys and prevents excessive loss of water from the body.

ADH per forms this activity by increasing the permeability of the walls of the renal tubule to reabsorb more water.

iv) Collecting duct

Collecting duct is where the urine is collected. The urine consists of

- ✓ **urea**
- ✓ **excess water**
- ✓ **salt** that are not reabsorbed back into the blood

finally the reabsorbed substances in the capillaries move in to venuoles (smallest veins) and then to **renal vein**. By the renal Vein, they are carried in to **inferior Venacava** which returns or carries purified blood back in to the heat.

Osmotic pressure of the blood and extracellular fluids are monitored by specialized cells in the brain called **osmo receptors**.

- ✓ If the water (osmotic) content of the blood is lower osmoreceptors stimulate the release of ADH by pitutary gland
- ✓ If the water content of the blood is too high the amount of ADH produced is less. However, if the concentration of blood is less more ADH is secreted to reabsorb more water.

Table: summary of what happens in the nephron

Proximal tubule	Loop of Henle	Distal tubule
Reabsorption of: ✓ 100% of glucose molecule ✓ 80% of water by Osmosis ✓ 67% of sodium ions	- further reabsorption of water - formation of urine - maintain ace of PH balance of the blood	- water balance by ADH - reabsorption of salt

Illustrative Examples

138. Which of the following is NOT the function of the kidney?
A. Deamination
B. Osmoregulation
C. Excretion
D. Water balance

Explanation: Osmoregulation (water balance) and excretion are the functions of the kidney. While, deamination (removal of excess omino acids from the body) is the function of the liver.

Answer: A

139. Why do we produce less urine during hot dry weather than during cold weather for the same amount of water we drink?
A. Because we produce less amount of waste.
B. Because our metabolic process is low.
C. Because much water is lost through sweat.
D. All of the above are correct.

Explanation: In hot weather much water is lost from the body through the sweat. That is why we produce less amount of urine in such a condition.

Answer: C

140. In male humans, which of the following is correct route of urine flow?
A. Bladder → Ureter → Urethra
B. Urethra →Bladder→ Ureter
C. Ureter → urethra →Bladder
D. Ureter →Bladder→Urethra

Explanation: The correct route of urine flow is:
Ureter → bladder → urethra

Answer: D

141. Which of the following is NOT part of human kidney?
A. Glomerulus
B. Pyramid
C. Medulla
D. Urethra

Explanation: Urethra is not part of the kidney. It is a part of a urinary system that carriers urine out of the body.

Answer: D

142. The tube that brings urine from kidney to urinary bladder is:
A. Renal artery
B. Glomerulus
C. Ureter
D. Urethra

Explanation: Ureter is a tube that carriers urine from kidney to urinary bladder.

Answer: C

143. Which of the following is not part of the nephron?
A. Bowman's capsule
B. Loop of Henle
C. Glomerullus
D. First convoluted tube
E. None of the above

Explanation:Nephron is the smallest unit of the kidney that has the following parts:

- ✓ Bowman's is capsule
- ✓ Loop of Henle
- ✓ Glomerulus
- ✓ First convoluted tubule
- ✓ Distal convoluted tubule
- ✓ Collecting duct

Answer: E

144. Of the following the substance which is present in the glomerular filtrate but normally completely reabsorbed as the filtrate passes along the tube is:

A. Glucose
B. Protein
C. Sodium ion
D. Urea

Explanation: Glucose is 100% reabsorbed from glomerular filtrate which passes along the renal tubules.

Answer: A

145. Which of the following is the principal organ for maintaining the body's water homeostasis?

A. Skin
B. Lungs
C. Kidneys
D. Liver

Explanation: Skin, lung and kidneys play role in water balance. However, the principal organ where most of the osmoregulation (water balance) occurs is the **kidney**.

Answer: C

146. Filtrate to from glomerulus first moves into:

A. Loop of Henle
B. Collecting duct
C. Proximal convoluted tubule
D. Distal convoluted tubule

Explanation: Filtrate from glomerulus first moves into proximal (first) convoluted tubule.

Answer: C

147. Which of the following structure of the urinary system connects the kidneys with the bladder?

A. Loop of henle
B. Renal vein
C. Urethra
D. Ureter

Explanation: Ureter is a tubule that connects kidneys with urinary bladder.

Answer: D

148. Which of the following shows the correct path of blood in a nephron?

A. Renal artery → glomerulus → capillaries → coiled tube → renal vein.
B. Renal vein →Renal artery → glomerulus → capillaries → coiled tube →renal artery

C. Renal artery → glomerulus → coiled tube → capillaries → renal vein

D. Renal vein → renal artery → coiled tube → capillaries → renal artery

Explanation: The correct path of blood in the nephron is: Renal artery → glomerulus → coiled (renal) tube → capillaries **Answer: C**

149. In which part of the nephron does ultrafiltration takes place?
A. Bowman's capsule
B. Distal convoluted tubule
C. Collecting ducts
D. Loop of Henle

Explanation: Ultra (pressure) filtration takes place in a part of a nephron known as Bowman's capsule. **Answer: A**

150. Which of the following is NOT found in the glomerular filtrate?
A. Salt
B. Glucose
C. Plasma protein
D. Water

Explanation: Glomerullar filtrate consists of small sized particles such as glucose, water, mineral, salt, urea. Large size particles such as blood cells and blood proteins are not found in glomerular filtrate. **Answer: C**

151. A man may drink several liters of water for some days and very much less for the other days. However, the water balance of his body is always maintained. Which of the following organs of the body is responsible for such physiological process?
A. Kidney
B. Heart
C. Lung
D. Liver

Explanation: Kidney is an organ that is involved in maintaining the water balance (osmoregulation) of the body. **Answer: A**

152. The human urinary system consists of:
A. A pair of kidneys, a pair of uterus and a bladder.
B. A kidney, a pair of urethra, a bladder.
C. A pair of kidneys, a pair of urethra and a bladder.
D. A pair of kidneys, a urethra, and bladders.

Explanation: A human urinary system consists of:
- ✓ A pair of kidneys
- ✓ A pair of ureter
- ✓ Urinary bladder
- ✓ Urethra

Answer: A

153. Which of the following does not normally happen as the glomerular filtrate is passing down the nephron tube?
A. Water is reabsorbed into the circulatory system.
B. Protein concentration increases.
C. Glucose concentration decreases.
D. Ammonium ion concentration increases.

Explanation: As a glomerular filtrate is passing down the nephron:
✓ most of the water is reabsorbed back into circulatory system.
✓ glucose concentration decreases because all the glucose molecules are reabsorbed back. **Answer: B**

154. In human urinary system, the temporary storage site of urine is:
A. Ureter C. Bladder
B. Urethra D. Nephron

Explanation: Urinary bladder is a part of urinary system where urine is temporary stored. **Answer: C**

155. A hormone released by pituitary gland to regulate water balancing process in the kidney is:
A. TSH B. FSH C. LH D. ADH

Explanation: Antiduretic hormone (ADH) is a hormone that is produced by pituitary gland when the amount of water in the body is too little. ADH, increases the permeability of capillaries surrounding thermal tube and make them reabsorb more water. **Answer: D**

156. The release of urine from a bladder is controlled by:
A. Urethra C. Ureter
B. Sphincter muscle D. Collecting duct

Explanation: Sphincter muscles are voluntary muscles found at the base of urinary bladder. They control the release of urine from the body. **Answer: B**

Critical thinking

Renal artery is a blood vessel that carries blood to the kidney and **renal vein** in a blood vessel takes blood from the kidney to inferior venacava. Therefore, which blood do you think is the purest? The one is renal vein or renal artery?

Liver as a Homeostatic Organ

Liver is reddish brown organ located in the abdomen under the diaphragm. It is a very special organ which plays a very important role in maintain the homeostatic stability of the body.
Liver has a special blood supply from the gut through **hapatic portal vein**.
Large number of reactions takes place in the liver. Most of these reactions are homeostatic. Some of these functions of the liver are:

1. Regulation of blood sugar level, since it is the storage site of glycogen.
2. Controlling and balancing fat and cholesterol levels in the blood.
3. Removal of excess amino acids from the body. This process is known as **deamination**.

> **Deamination** is the removal of the amino group of the amino acids. This occurs because our body cannot store excess protein.

4. Distinigration of worn out red blood cells.
5. Storage of hemoglobin.

> **Hemoglobin** is a red pigmented substance in the blood which is made up of substances such as ion. It is involved in transportation of oxygen in the body. Deficiency of iron causes **anemia**.

6. **Detoxification** – the process of removing or neutralizing toxic substances. People may get toxified or exposed to toxic substances by manmade or natural factors. Thus, the liver is an organ of the body which copes with toxic substances. For instance, people who consume excessive amount of alcohol (alcoholic people) mostly suffer from liver damage which may lead to death. Can you explain why is this so?
7. **Temperature regulation** – since many reactions take place in the liver, it produces/generatesmore heat for the body.

Illustrative Example

157. Which of the following is the function of the liver?
A. Excretion of excess salt
B. Excretion of urea
C. Deamination of amino acids
D. Maintaining water balance

Explanation: Liver is an organ that has several functions in the body. Among the given alternatives liver is involved in deamination of amino acids.

Answer: C

158. Which of the following process do not take place in the liver?
A. Deamination
B. Storage of glycogen
C. Detoxification of poison
D. Secretion of insulin and glycogen

Explanation: Liver is involved in
- ✓ Deamnaiton
- ✓ Storage of glycogen
- ✓ Detoxification of poison

Answer: D

159. The two functions of the liver are:
A. Secretion of bile and storage of glycogen
B. Secretion of hepatin and storage of proteases.
C. Secretion of insulin and storage of hapatin
D. Storage of bile and glycogen

Explanation: Liver secretes bile and stores glycogen.

Answer: A

160. Which of the following excretory product is made in the liver?
A. Minerals
B. Urea
C. Amino acids
D. Glucose

Explanation: Urea is the excretory product made in the liver as a result of deamination of excess amino acids.

Post Test – 13

Multiple Choice

1. Which of the following is NOT excretory product?
A. Urea B. Sweat C. Carbon dioxide D. Feaces

2. The excretory product which is produced in the lung is:
A. Urea
B. Caron dioxide
C. Salt
D. Ammonia

3. The largest proportion of human sweat is:
   A. Salt
   B. Water
   C. Nitrogenous wastes
   D. Urea

4. The correct path of urine flow in human urinary system is:
   A. Urethra → Bladder → Ureter
   B. Bladder → Ureter → Urethra
   C. Ureter → bladder→Urethra
   D. None of the above

5. Which of the following is NOT a glomerular filtrate?
   A. Urea
   B. Blood protein
   C. Water
   D. Salt

6. One of the following is NOT an excretory organ, which one is it?
   A. Heart
   B. Brain
   C. Kidney
   D. Liver

7. The hormone produced by pitutary gland to regulate water balance by the kidney is:
   A. Luteinizing hormone
   B. Follicle stimulating hormone
   C. Antidiuretic hormone
   D. Follicle stimulating hormone

8. The process by which a liver removes excess amino acids from the body is called:
   A. Detoxification
   B. Deamination
   C. Excretion
   D. Defecation

9. The pH balance of the blood is maintained in which part of renal tubule:
   A. Bowman's Capsule
   B. Loop of Henle
   C. Distal tubule
   D. Proximal tubule

10. Which one of the following is NOT a function of the liver?
   A. Osmoregulation
   B. Detoxification
   C. Deamination
   D. Sugar regulation

11. The re-absorption of salts during urine formation is allowed by
   A. Descending limp of loop of henle
   B. Ascending limp of loop of Henle
   C. Bowman's capsule
   D. Glomerulus

Supplementary Exercise

1. The end of fertility in the reproductive cycle of a human fem known as:
   A. Adolescence
   B. Menstruation
   C. puberty
   D. Menopause
2. Which of the following contraceptive methods results sterilization of males?
   A. Vasectomy
   B. Tubectomy
   C. Condom
   D. Norplant
3. The seminiferous tubules are part of the:
   A. Cell
   B. Kidneys
   C. testes
   D. ovaries
4. Which of the following is the functional unit of the kidneys?
   A. Ureter
   B. Bladder
   C. Nephron
   D. Urethra
5. The mammalian eye focuses by:
   A. Opening and closing the pupil
   B. Changing the shape of the eye
   C. Moving the lens forward and back
   D. Changing the shape of the lens.
6. The part of the ear that has cells sensitive to sound waves is:
   A. Tympanic membrane
   B. Cochlea
   C. semicircular canal
   D. pinna
7. Which of the following eye defects can be corrected by wearing eye glasses with convex lens?
   A. Long sight
   B. Blindness
   C. short sight
   D. Astigmatism
8. Which of the following organs forms urea?
   A. Lung
   B. Skin
   C. kidney
   D. liver
9. The part of the eye that is not involved in refraction of light rays is:
   A. Corenea
   B. Vitreous humour
   C. fovea
   D. lens
10. Which of the following hormone controls body metabolism?
   A. Thyroxin
   B. Adrenalin
   C. corticoid
   D. antiduretic hormone

11. The mylean sheath which wrap around the axons of some neurons is formed by:
    A. Nodes of Rain vier
    B. Dendrites
    C. synapses
    D. Schwann cells
12. Which part of the eye is pigmented and contains network of blood vessels that nourish the cells of the retina?
    A. Sclerotic layer
    B. Cornea
    C. chroid layer
    D. conjuctiva
13. The sensory neurons, stimuli are received by the:
    A. Dendrites
    B. Axons
    C. cell body
    D. Mylen
14. The amount of light entering the eye is determined by the size of the:
    A. Retina
    B. Pupil
    C. cornea
    D. fovea
15. Which of the following are the main components of urine in addition to water?
    A. Amino acids and fatty acids
    B. Urea and salts
    C. Ammonia and bile
    D. Hydrochloric acid and bases
16. When you look at an intact human brain, what you see is a large highly convoluted outer surface. This is are:
    A. Cerebral cortex
    B. Medulla
    C. cerebellum
    D. reticular system
17. Identify the correct sequence of structures through which glomerular filtrate passes on its journey through the nephron?
    A. Bowman's capsule, proximal tubule, loop of Henle, distal tubule, collecting duct.
    B. Loop of Henle, proximal tubule, Bowman's tubule, distal tubule, collecting duct.
    C. Bowman's capsule, collecting duct, loop of Henle, proximal tubule, distal tubule.
    D. Loop of Henle, Bowman's capsule, distal tubule, proximal tubule, collecting duct.
18. Which of the following diseases is caused by the deficiency of thyroxin that results in reduction of mental and physical development in infants?
    A. Goiter B. Gigantism C. cretinism D. obesity

A complete human gestation period occurs between the times of:
A. Blastula formation and implantation
B. Fertilization and birth
C. Ovulation and cleavage
D. Birth and puberty

After fertilization, the resulting zygote begins to divide by:
A. Syngamy B. Binary fussion C. mitosis D. meiosis

Which of the following structures of a mammalian eye does not refract light?
A. Pupil B. Vitreous humour C. cornea D. retina

The correct path of nerve impulse in a reflex arc is:
A. Effector → sensory neuron → association neuron → receptor→ motor neuron
B. Receptor → motor neuron→ association neuron → sensory neuron→ receptor
C. Effector → sensory neuron → motor neuron→ association neuron → receptor
D. Receptor → sensory neuron → association neuron → motor neuron→ receptor

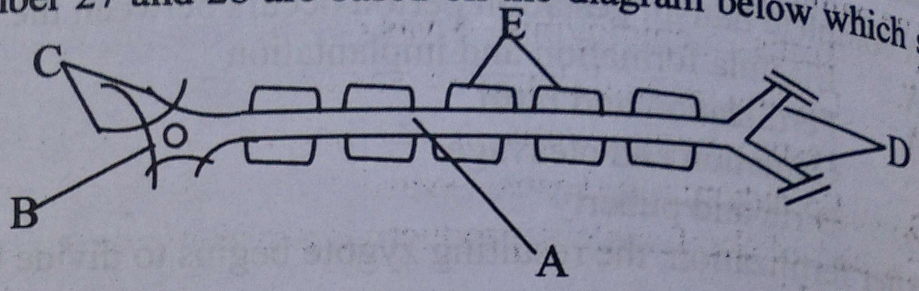
Which of the following eye problems is caused by poor diet?
A. Trachoma C. astigmatism
B. Night blindness D. long sight

The eye lens is flexible in shape. This allows to:
A. Control the amount of incoming light
B. Accommodate near and distant objects
C. Transmit different colors of light
D. Pass more light in dim condition

Which of the following hormones controls the functions of endocrine glands?
A. Adrenaline C. Tropic hormone
B. Prolactin D. thyroxine

Which of the following is true of a resting neuron?
A. Its outside membrane becomes negative
B. It is polarized
C. It has an action potential
D. It is depolarized

Item number 27 and 28 are based on the diagram below which shows a neuron.



27. The part of a neuron indicated by "A" is:
    A. Dendron
    B. Axon
    C. dendrite
    D. terminal branch
28. This neuron is a/an:
    A. Interneuron
    B. Motor neuron
    C. sensory neuron
    D. association neuron
29. The axon of motor neuron is connected to:
    A. Muscle fibers
    B. Dorsal root
    C. sensory receptors
    D. association neurons
30. If a mother decides to avoid pregnancy by using arhythm method and if her menstruation begins on sene 12, she should avoid sexual intercourse in the days between:
    A. Hamle 10 – 18
    B. Sene 30 – Hamble 8
    C. sene 23 – Hemle 2
    D. sene 14 – sene 20
31. Which of the following is not correct about the pancrease? It produces,
    A. Glucagon
    B. Fats emulsifying salts
    C. Insulin
    D. digestive enzymes
32. Which of the following substance is abundant in urine?
    A. Amino acids
    B. Urea
    C. Glucose
    D. sodium
33. Which of the following functions of the liver is related to the hormone glucagon?
    A. Detoxification of alcohol and drugs
    B. Production of bile salts
    C. Conversion of glycogen into glucose
    D. Damnation of excess amino acids
34. The dendrites of a neuron can carry an impulse only.
    A. Toward the brain
    B. Away from the cell body
    C. Away from the spinal cord
    D. toward the cell body

The pathway of a sperm in male reproductive structures is:
A. Testis → epididymis → vas deferens → ureter
B. Urethra → epididymis → vas deferens → testis
C. Testis → epididymis → vas deferens → urethra
D. Epididymis → Testis →vasdefens→ urethra

The part of the female reproductive organ where the embryo is implanted is:
A. Cervix B. Ovary C. uterus D. Oviduct

In dim light
A. Pupil dialates
B. Radial muscles contract
C. Circular muscle contract
D. Iris flattens

The eye defeact which is corrected by wearing equally curved cylindrical lenses is:
A. Cataract C. short sight
B. Long sight D. Astigmatism

The region in the brain which is concerned with vision is:
A. Parietal lobe C. occipital lobe
B. Optic lobe D. eye lobe

Which of the following sense organs have chemo receptors?
A. Eye and nose C. skin and tongue
B. Ear and tongue D. Nose and tongue

The newly discovered sense of taste is called.
A. Sweat C. Bitter
B. Umami D. Salts

If someone got angry, the adrenal gland is stimulated and produced the adrenaline hormone. Which of the following does not happen:
A. Decline in metabolic rate
B. Dialation of pupil
C. Blood sugar level increases
D. Heart beat rate raises

After taking some coffee, the body of to person become stimulated. What is the name of the drug that stimulated
A. Khatonine C. caffeine
B. Cocaine D. heroin

44. Which of the following drug is illegal in Ethiopia?
    A. Khat
    B. Cigarette
    C. Heroin
    D. alcohol

45. The part of the eye ball which nourishes the cells of the retina is:
    A. Chroid
    B. Sclera
    C. pupil
    D. lens

46. In human female on the average the egg is released from the ovary around the
    A. 14^{th} day of the menstrual cycle
    B. 28^{th} day of the menstrual cycle
    C. 5^{th} day of the menstrual cycle
    D. 2^{nd} day of the menstrual cycle

47. The development of the female secondary sexual characteristics is due to:
    A. Androgen
    B. Oestrogens
    C. testosterone
    D. thyrotrophin

48. The intrauterine device (IUD) contraception method prevents pregnancy by
    A. Hindering sperm from passing to fallopian tube
    B. Stopping the release of egg from the ovary
    C. Blocking the passage of hormones to the uterus
    D. Hindering implantation of fertilized egg to the uterine wall

49. In the human female, on the average, Ovulation takes place
    A. In the first three days after menstruation
    B. In the last four days before menstruation
    C. Simultaneously with ovulation
    D. About half – way between two menstruations
    E. Ovulation and menstrual cycle have no relationship

50. Which of the following process decreases when the human body temperature decreases?
    A. Secretion of sweat
    B. Blood flow to the internal organs.
    C. Shivering
    D. Secretion of insulin

51. The pituitary gland release more ADH in to the kidney tubules when;
    A. The water level of the body is too high
    B. The water level of the body is too little
    C. The water level of the body is constant
    D. It has no relationship with the water level of the body.

52. Specialized cells in the brain which are responsible for monitoring the osmotic pressure is:
A. Osmosis C. osmo receptors
B. Osmo regulation D. osmotic uptake

53. The most reliable contraceptive method is:
A. Condom B.Injectables C. IUD D. Pills

54. The most effective contraceptive method is:
A. Sterilization C. IUD
B. Rhythmethod D. diaphragm

55. The correct order of ossicles from inner to outer ear is:
A. Stapes → malleus → Incus
B. Stapes → Incus → malleus
C. Malleus → Incus → Stapes
D. Incus → malleus → Stapes

56. The part of the retina which does not contain light sensitive cells is:
A. Yellow spot C. Aqueous humor
B. Blond spot D. vitreous humor

57. In a deporized neuron?
A. The outside of axon membrane is positively charged
B. The inner side of axon membrane is positively charged
C. The inner side of axon membrane is negatively charged
D. There is no charge deference across the axon membrane.

58. The main excretory organ of the body is:
A. Skin B. Lung C. kidney D. liver

59. The cells of the retina that are sensitive to bright light are:
A. Cone B. Rod C. chroid D. sclera

60. The refraction of light in the eye ball is prevented by:
A. Vitreous humor B. Pupil C. chroid D. sclera

Fill in the Blank

61. A fatty insulating layer around the axon is called _____.
62. _____ is a wave of depolarization across the neuron.
63. The central envious system consists of _____ and _____.
64. _____ is a neural pathway during reflex action.
65. A sudden, automatic and uncontrolled response of the body to stimuli is called _____.
66. A stimulant drug found in khat is _____

67. _____ is the abity of the eye to see near and distant objects.
68. The male gonad is called _____, while, the fenale gonad is called _____.
69. The part of the Female reproductive organ where fertilization takes place is called _____.
70. The end of fertility in women which occurs around the age of 45 to 50 is called _____
71. The sterilization of male is called _____, while the sterilization of Females is called _____.
72. The process of maintaining constant internal condition of the body is called _____.
73. Animals with variable body temperature are called _____.
74. _____ is the widening of blood vessels under the skin which results in heat loss from the body.
75. The process of regulating the water balance and ionic control of the body is called _____
76. _____ are clusters of capillaries in the Bowma's capsule.
77. The process of removing excess amino acids from the body is called _____.

Answers to Supplementary Exercise

1.	D	16.	A	31.	B	46.	A
2.	A	17.	A	32.	B	47.	B
3.	C	18.	C	33.	C	48.	D
4.	C	19.	B	34.	D	49.	D
5.	D	20.	C	35.	C	50.	A
6.	B	21.	D	36.	C	51.	B
7.	A	22.	D	37.	B	52.	C
8.	D	23.	B	38.	D	53.	D
9.	C	24.	B	39.	B	54.	A
10.	A	25.	C	40.	D	55.	B
11.	D	26.	B	41.	B	56.	B
12.	C	27.	B	42.	A	57.	B
13.	A	28.	B	43.	C	58.	C
14.	B	29.	D	44.	C	59.	A
15.	B	30.	C	45.	A	60.	C

69. Mylensheath
70. Action potential
71. Brain, spinal cord
72. Reflex arc
73. Reflex action
74. Khatonine
75. Accomodation
76. Testis, ovary
77. Oviduct
78. Menopause
79. Vasectomy, tubectomy
80. Homeostasis
81. Piokilotherms
82. Vasodialation
83. Osmoregulation
84. Glomerullus
85. Deamination

Answers to Post Test 1

Part – I

1.	B	2.	C	3.	A	4.	B	5.	B
6.	A	7.	B	8.	B	9.	B	10.	B
11.	C	12.	C						

Answer to Post Test – 2

Part – I

1.	C	2.	C	3.	C	4.	A	5.	A	6.	A
7.	B	8.	C	9.	A	10.	B	11.	A	12.	B

Answer to Post Test – 3

Part – I

1.	D	2.	B	3.	B	4.	C	5.	B
6.	C	7.	B					

Answers to Post test – 4

Part – I

1.	D	2.	A	3.	A	4.	C	5.	B
6.	B	7.	A	8.	A	9.	C	10.	D

Part – II

1.	C	2.	B	3.	A	4.	D

Answers to post test 5

Part – I

1.	D	2.	D	3.	B	4.	B	5.	A
6.	C	7.	B	8.	A	9.	C	10.	B

Part – II

1. Cornea 2. Cone, rod 3. Iris 4. Optic nerve
5. Accommodation

Part – III

1.	m	2.	K	3.	I	4.	L	5.	F
6.	A	7.	g	8.	J	9.	B	10.	D
11.	C	12.	H						

Answer to Post Test – 6

Part – I

1.	C	2.	D	3.	C	4.	B	5.	C
6.	A	7.		8.	C	9.	A	10.	B
11.	A	12.	A						

Part – II

1.	B	2.	E	3.	C	4.	D	5.	A

Answer to Post Test – 7

Part – I

1.	B	2.	C	3.	D	4.	B	5.	C

Part – II

1.	C	2.	B	3.	D	4.	A		

Answers to Post Test – 8

Part – I

1.	C	2.	A	3.	B	4.	A	5.	C

Part – II

1.	D	2.	E	3.	A	4.	B	5.	C

Answers to Post Test – 9

Part – I

1.	B	2.	C	3.	A	4.	B	5.	
6.	B	7.	D	8.	B	9.	B	10.	A
11.	A	12.	D	13.	B				

Answers to Post Test – 10

Part – I

1.	D	2.	C	3.	C	4.	A	5.	D
6.	B	7.	C	8.	A	9.	B	10.	B
11.	A	12.	B						

Answers to Post Test – 11

Part – I

1.	B	2.	A	3.	D	4.	B	5.	C
6.	ii, iv, iii, i, v								

Answers to Post Test – 12

Part – I

1.	D	2.	D	3.	B	4.	D	5.	B
6.	D	7.	B	8.	C	9.	B	10.	C
11.	C	12.	A	13.	B	14.	D	15.	D
16.	C	17.	C	18.	B	19.	D		

Answers to Post Test 13

Part – I

1.	D	2.	B	3.	B	4.	C	5.	B	6.	B
7.	C	8.	B	9.	B	10.	A	11.	A		

Unit – 10

10. Food Making and Growth in Plants

Introduction

Plants are groups of organisms that have the following features in common. They:

- ✓ are photosynthetic (photo artsthrophs)
- ✓ are eukaryotic and multicellular

Plants can be classified into two major categories. These are:

- ✓ **Lower plants** – are small sized plants that do not have well-structured parts. Such as root, leaf and stem.
  Examples: Mosses, Ferns, liverworts
- ✓ **Higher plants** – are groups of plants that posses the major parts of a plant such as root, leaf and steam.
  Examples: Angiosperms, gymnosperms

The major parts of higher plants have the following major functions.

- ✓ **Leaves** – are photosynthetic organs that are capable of converting light energy, carbondioxide and water into organic substances.
- ✓ **Stem** – is a part which provides support and transportation of substances. Such as water, minerals and food.
- ✓ **Root** – anchors the plant to the ground and involved in the absorption of water and minerals.
- ✓ **Flowers** – are the sexual reproductive organs of flowering plants.

10.1 The Leaf

Leaf is an organ of a plant where the process of photosynthesis takes place. Thus, it is also known as a photosynthetic machine of a plant.

Photosynthesis: is the process by which plants and some bacterias manufacture their own food from the sunlight energy.

Parts of a leaf

The parts of a leaf can be seen from external and internal sections.

The external structures of a Leaf

The external part of a leaf has the following major components. These are:

- ✓ **Blade** – is an extended surface of the leaf. It is broader in some plants and narrower in some other groups of plants.
- ✓ **Petiole** – connects the leaf to the stem.
- ✓ **Midrib** – is the main vein and it also consists of vascular tissues (xylem and phloem tissues)
- ✓ **Margin** – is the border or the edge of the leaf.
- ✓ **Apex** – is the tip of the leaf.

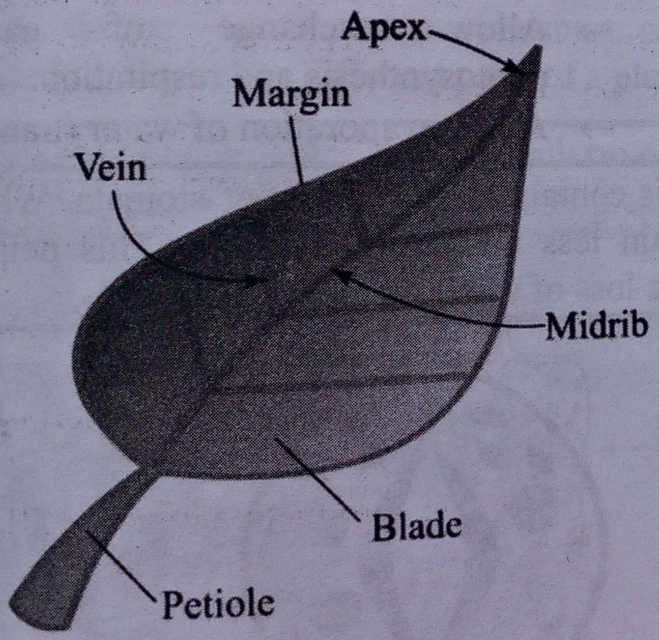


Figure 10.1: The external structure of a leaf

The internal Structure of the leaf

The internal structure or the cross sectional area of the leaf has the following major parts. These are:

- ✓ Upper and lower epidermis
- ✓ Mesophyll layers
- ✓ Vascular bundles

i) Epidermis

Epidermis is the most outer layer of the leaf. It is composed of epidermal cells that do not have chloroplast.

- ✓ The upper epidermis is covered with a tin plastic like layer known as **waxy cuticle**. Waxy cuticle prevents excessive loss of water by transpiration.

Transpiration – is evaporation of water from a plant mainly through a leaf.

- ✓ The lower layer of epidermis are modified into bean shaped cells known as **guard cells**. Guard cells contain chloroplast. Thus, they are photosynthetic.
  - o The opening between two guard cells makes a space that is known as **stomata (singular stoma)**. Stomata opening have the following key importance's:
    - → Allow exchange of gases during photosynthesis and respiration.
    - → Allow evaporation of water (transpiration).

Lower epidermis contain large number of stomata. While the upper epidermis contain less number of stomata. This helps a plant to reduce excessive loss of water by transpiration.

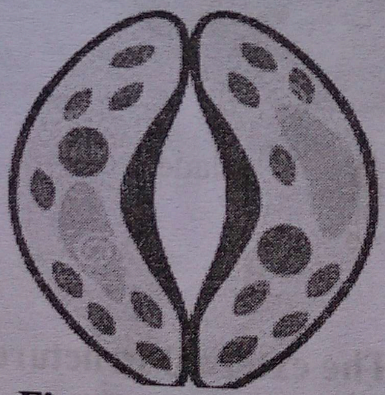


Figure 10.2: Guard cells

Mesophyll layers

The mesophyll layer has two major types of tissues. These are:

- ✓ **Palisade mesophyll** – are vertically arranged and closely packed cells. The Palaside cells contain the largest number of chloroplasts in the leaf. Thus, they are the main site of photosynthesis is in the leaf.

- ✓ **Spongy mesophyll** – are rounded cells that are a little apart from one another. This creates a big surface area for exchange of gas in the leaf.
  - They also contain chloroplasts, even though it is not as much as that of palisade cells. Thus, spongy cells are also photosynthetic.
  - Spongy meslophylls serve mainly for the movement of carbondioxide needed for the process of photosynthesis.

Vascular bundles

Vascular bundles contain tissues that serve for transportation of different substances. These are:

- ✓ **Xylem tissue**: used to transport water from the root to the leaves. This water is a raw material for the process of photosynthesis that takes place in the leaf.
- ✓ **Phloem tissue** – are used to transport organic substances produced by the leaf to different parts of a plant.

Chloroplasts are the organelles in the cytoplasm of plant cells where chlorophyll is stored and photosynthesis takes place.

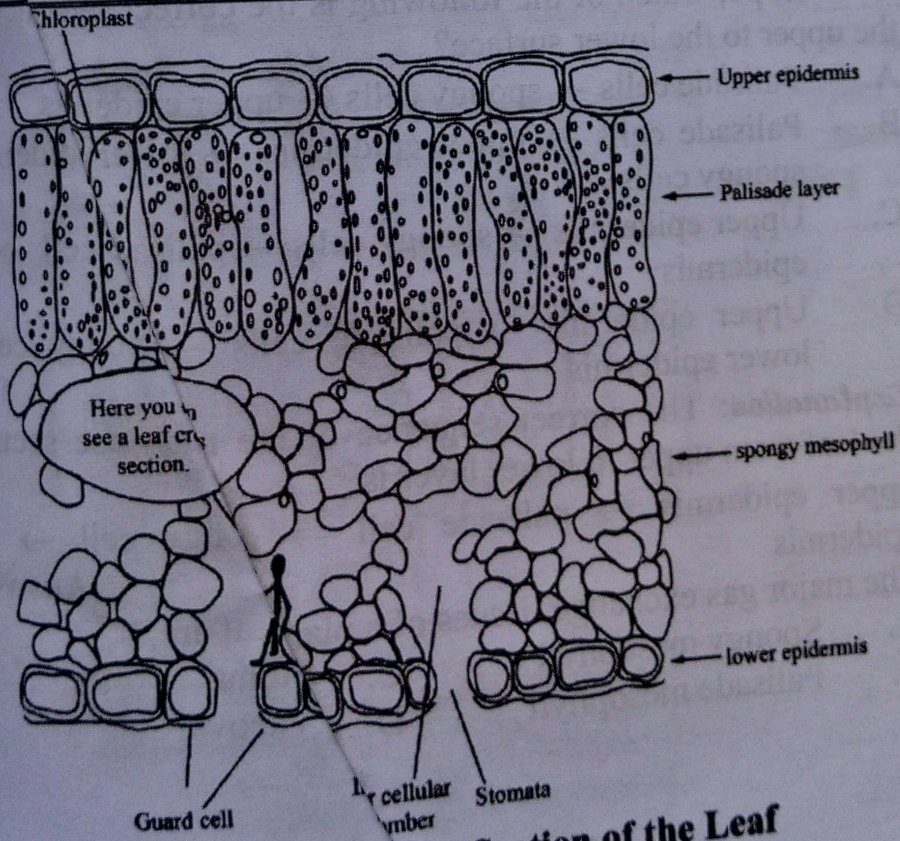


Figure 10.3: Cross Section of the Leaf

Illustrative Example

1. The function of the epidermal cells of a leaf is to:
   A. Undergo photosynthesis
   B. Stored food
   C. Minimize water loss
   D. Absorb light

   Explanation: Epidermal cells contain waxy cuticles that prevents excessive loss of water by transpiration. **Answer: C**

2. Which of the following plant cells do not photosynthesize?
   A. Palisade cells
   B. Phloem cells
   C. Guard cells
   D. Spongy cells

   Explanation: Phloem cells do not contain chlorophyll. Thus, they do not photosynthesize. **Answer: B**

3. Mesophyll layer of a leaf has two distinct layers known as:
   A. Upper and lower epidermis
   B. Spongy and epidermis
   C. Palisade and spongy
   D. Palisade and air space

   Explanation: Mesophyll layer is the middle portion of the transverse section of the leaf. It consists of **palisade** and spongy layers. **Answer: C**

4. When you look at the transverse section of a leaf under a microscope, which of the following is the correct sequence, from the upper to the lower surface?
   A. Palisade cells → spongy cells → upper epidermis
   B. Palisade cells → upper epidermis → lower epidermis → spongy cells
   C. Upper epidermis → spongy cells → palisade cell → lower epidermis
   D. Upper epidermis → palisade cells → spongy cells → lower epidermis

   Explanation: The correct sequence of the transverse section of the leaf from upper to lower layer is:
   upper epidermis → palisade cell → spongy cell → lower epidermis **Answer: D**

5. The major gas exchange tissues of a plant leaf are
   A. Spongy mesophyll
   B. Palisade mesophyll
   C. Stoma
   D. Vacovle

Explanation: Spongy cells are rounded cells that are arranged far apart from one another. They serve as gas exchange and photosynthetic layer in a leaf. **Answer: A**

6. Which of the following parts of a dicot plant leaf contains the largest number for chloroplasts?

A. Palisade layer C. Epidermis
B. Spongy layer D. Stomata

Explanation: The largest number of chloroplast is found in palisade mesophyll. Thus, palisade cells are the most photosynthetic cells in a leaf. **Answer: A**

7. Which structure of a leaf prevents excessive loss of water from the leaf surfaces?

A. Spongy water C. Air spaces
B. Palisade layer D. Cuticle

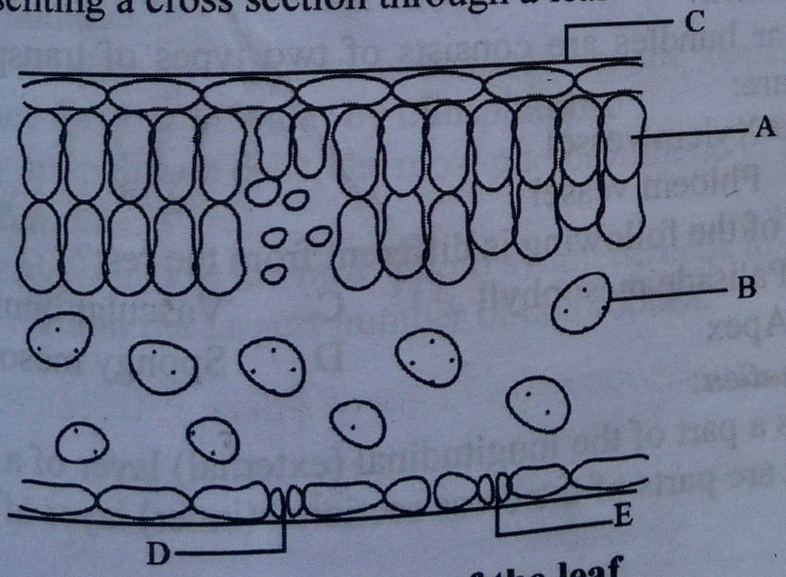
Explanation: Waxy cuticle is structure that is found on the surface of upper epidermis. It prevents excessive loss of water from the leaf surfaces. **Answer: D**

8. Which of the following parts of a leaf is known as main vein?

A. Apex B. Midrib C. Petiole D. Blade

Explanation: Midrib is a structure located on the external surface of the leaf. It is also known as the main vein. **Answer: B**

Examples 9 and 10 are based on the following diagram representing a cross section through a leaf.



Cross Section of the leaf

9. Structures 'A' and 'B' represent, respectively.
   A. Palisade layer and air spaces
   B. Spongy layer and air spaces
   C. Palisade and air spaces
   D. Spongy layer and chlorophyll

Answer: A

10. Structures 'D' and 'E' represent, respectively:
   A. Stomata and guard cell
   B. Epidermal cell and stomata
   C. Waxy cuticle and stomata
   D. Guard cell and stomata

Answer: D

11. The most dominant plants on earth at present are:
   A. Lycophytes
   B. Angiosperms
   C. Bryophytes
   D. Gymonosperms

Explanation:
The most dominant and diverse groups of plants on the earth are flowering plants or angiosperms.

Answer: B

12. Why is the lower epidermis contain large number of stomata than the upper epidermis:

Explanation: The number of stomata is leaser on the upper epidermis as compared to the lower epidermis. This helps a plant to reduce water loss by transpiration.

13. Vascular bundles are composed of:
   A. Palisade and spongy layer
   B. Xylem and phloem vessels
   C. Xylem and palisade layer
   D. Phloem and spongy layer

Explanation:
Vascular bundles are consists of two types of transport tissues. These are:
   - ✓ Xylem vessel
   - ✓ Phloem vessel

Answer: B

14. Which of the following is different from the rest?
   A. Palisade mesophyll
   B. Apex
   C. Vascular bunles
   D. Spongy mesophyll

Explanation:
Apex is a part of the longitudinal (external) layer of a leaf. While, the rest are parts of the cross sectional (inner) layer of the leaf.

Answer: B

15. Which one of the following is incorrectly matched with its function?
A. Palisade mesophyll – photosynthetic layer
B. Vascular bundles – transportation of substances
C. Waxy cuticle – increases transpiration rate
D. Spongy mesophyll – serves as air space

Explanation: Waxy cuttle reduces or minimizes the rate of transpiration.

Answer: C

16. Describe the functions of the following parts of a leaf
- ✓ Waxy cuticle
- ✓ Stomata
- ✓ Vascular bundles
- ✓ Palisade mesophyll
- ✓ Spongy mesophyll

Explanation:

Parts of a leaf	Function
Waxy cuticle	Prevents evaporation or loss of water
Stomata	Exchange of gases
Vascular bundles	Transportation of substances (Food water and minerals)
Palaside mesophyll	Photosynthesis
Spongy mesophyll	Air space and photosynthesis

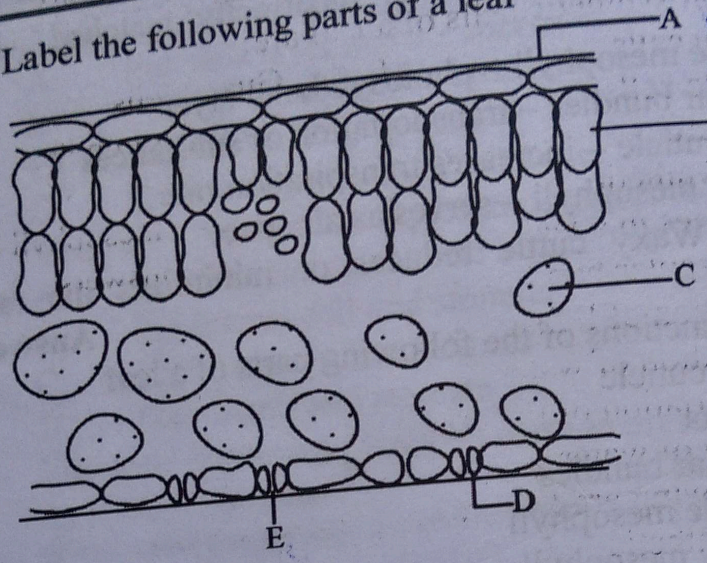
17. Why is the number of stomata opening reduced on the upper epidermis, as compared to the lower epidermis?

Explanation:
The number of stomata is reduced on the upper epidermis to reduce the loss of water by transpiration.

18. Why are palisade cells, the most photosynthetic in the leaf?

Explanation:
Palisade cells are the most photo synthetic cells in a leaf, because they contain the largest number of chloroplast.

19. Label the following parts of a leaf



Explanation:

- ✓ A = upper epidermis
- ✓ B = Palisade mesophyll
- ✓ C = spongy mesophyll
- ✓ D = Guard cell
- ✓ E = stomata

Post Test I

Part I: Choose the best answer from the given alternatives

1. Which one of the following is NOT the part of longitudinal section of a leaf?
   A. Petiole B. Blade C. Epidermis D. Lamina
2. Structures on the upper epidermis of the leaf that prevents excessive evaporation of water is:
   A. Waxy cuticle C. Guard cell
   B. Stomata D. Chlorplast
3. Evaporation of water through the leaf of plants is:
   A. Translocation C. Conduction
   B. Transpiration D. Convection
4. Large number of stomata are found at:
   A. Lower surface of epidermis
   B. Upper surface of epidermis
   C. Evenly distributed through all parts of epidermis
   D. Not found at both surfaces
5. The only cells of epidermis that contain chloroplasts are:
   A. Spongy cells C. Guard cells
   B. Palisade cells D. All

6. The most photosynthetic cells of a leaf are:
   A. Spongy cells
   B. Palisade cells
   C. Guard cells
   D. Vascular bundles

7. Spongy cells are
   A. Photosynthetic
   B. Air spaces in a leaf
   C. Parts of mesophyll layers
   D. All are answers

8. The role of vascular bundles in a leaf is:
   A. Absorption of light
   B. Evaporation of water
   C. Transportation of materials
   D. Splitting of water

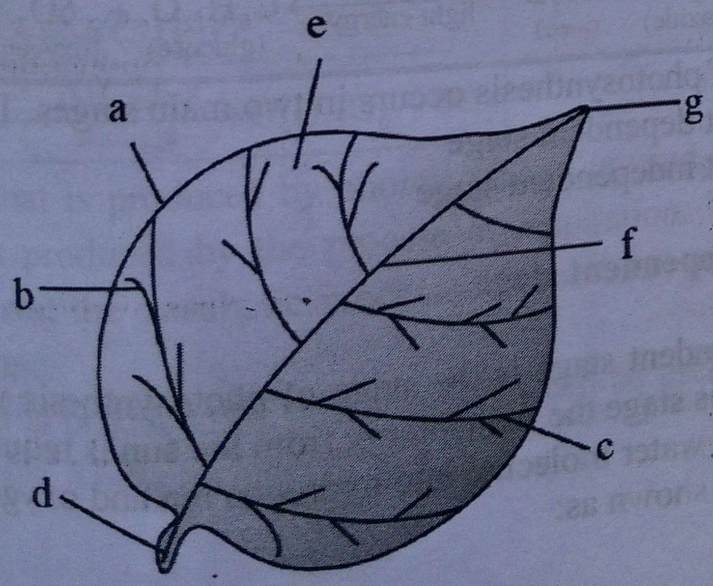
9. All of the following are features of plants, except:
   A. Photosynthetic
   B. Eukaryotic
   C. Multicellular
   D. Heterotrophs

10. Which are is NOT photosynthetic in a leaf?
   A. Palisade cells
   B. Spongy cells
   C. Guard cells
   D. Xylem cells

Part II: Write short answers for the following.

11. Rearrange the following parts of a leaf according to their sequential arraignment from upper to lower epidermis.
   - Spongy mesophyll
   - Upper epidermis
   - Lower epidermis
   - Palisade mesophyll

12. Label the following external parts of a leaf.



Part III: Match the following parts of a leaf with their function.

Column A	Column B
13. Xylem tissue	A. Controls the opening and closing of stomata
14. Palisade cells	B. Serve as air space in the leaf.
15. Guard cells	C. Transportation of water and minerals.
16. Waxy cuticle	D. Prevents excessive loss of water by transpiration
17. Spongy cells	E. The most photosynthetic layer in a leaf.

Photosynthesis

Photosynthesis is the process in which green plants synthesize organic substances (sugar) from the inorganic raw materials found in their surroundings.

For the process of photosynthesis to take place, the main requirements are:

- ✓ Chlorophyll is the light absorbing pigment found in the chloroplast.
- ✓ Water – from the soil
- ✓ Carbon dioxide – from the air
- ✓ Light – from the sun

The overall reaction of photosynthesis is summarized as:

$$\underset{\text{(carbondioxide)}}{6CO_2} + \underset{\text{(water)}}{6H_2O} \xrightarrow[\text{light energy}]{\text{chlorophyll}} \underset{\text{(glucose)}}{C_6H_{12}O_6} + \underset{\text{(oxygen)}}{6O_2}$$

The process of photosynthesis occurs in two main stages. These are:

- ✓ Light dependent stage
- ✓ Light independent stage

i) Light dependent stage

The light dependent stage is the stage of photosynthesis where light is required. At this stage the light energy from the sun is required to:

- ✓ Split water molecule into hydrogen ion and oxygen molecule.

The reaction is shown as:

$$12H_2O \xrightarrow[\text{chlorophyll}]{\text{sunlight}} 24H^+ + 6O_2$$

Among the products of the light dependent stage:

- ✓ The oxygen molecule is released into the atmosphere as a bi-product.
- ✓ The hydrogen ions are trapped by a hydrogen acceptor substance called $NADPH_2$ in order to be used in the light dependent stage.
- ✓ The remaining light energy is charged into a form of chemical energy called ATP, which serves as energy for the light independent stage.

The light dependent stage of photosynthesis takes place in the part of chloroplast called **grana**.

ii) The Light Independent Stage

The light independent stage of photosynthesis does not directly require light. But, this does not mean it occurs at the night time.

This stage involves the use of the products form light dependent stage and carbondioxide from the air to form sugar (glucose).

The reaction of light independent stage of photosynthesis is shown as:

$$24H^+ + 6CO_2 \rightarrow C_6H_{12}O_6 + 6H_2O$$

The light independent stage of photosynthesis takes place in the **stoma** of the chloroplast.

The overall reaction of photosynthesis is:

$$12H_2O \xrightarrow[\text{sunlight}]{\text{chlorophyl}} 24H^+ + 6O_2$$

$$24H^+ + 6CO_2 \longrightarrow C_6H_{12}O_6 + 6H_2O$$

$$6H_2O + 6CO_2 \xrightarrow[\text{sunlight}]{\text{chlorophyll}} C_6H_{12}O_6 + 6O_2$$

The glucose that is produced by photosynthesis is used as a source of energy that is produced by the process of respiration. This energy is used to carry out the metabolic processes of the body such as growth, reproduction, etc.

The sugar that is produced by photosynthesis can be used by the plants in different ways. These include:

- ✓ Built into starch for storage
- ✓ Built into molecules like fructose (fruit sugar), sucrose to be transported around the plant.
- ✓ Built up into molecules like **cellulose** to make new plant cell walls.
- ✓ Use to make **amino acids** (building blocks of proteins) along with nitrates and other nutrients.
- ✓ Built up into fat and oils (lipids) for storage, in seed and make part of cell membrane.
- ✓ Building important molecules such as **chlorophyll**.

Glucose is mainly stored in the form of starch which is stable and insoluble form of carbohydrate.

Illustrative Examples

20. The first stage of the reaction of photosynthesis does not involve:
    A. Light energy
    B. Carbondioxide
    C. Water
    D. Chlorophyll

    Explanation: The first stage of photosynthesis is the light dependent stage and in involves **light energy, water** and **chlorophyll.** **Answer: B**

21. Which of the following is true about photosynthesis?
    A. It takes place in green plants.
    B. It requires sunlight and oxygen.
    C. It releases carbon dioxide to atmosphere.
    D. It reduces the level of atmosphere oxygen.

    Explanation: The process of photosynthesis:
    - ✓ takes place in green plants
    - ✓ requires sun light and carbondioxide
    - ✓ releases oxygen to the atmosphere
    - ✓ reduces the CO_2 level in the atmosphere

    Answer: A

22. Which of the following leaf features is important for photosynthesis?
    A. Margin
    B. Size of Petiole
    C. Arrangement
    D. Venation

    Explanation: Leaf vennations are very important features for photosynthesis, because it contain xylem vessels through which water is supplied to the chloroplasts in the leaf. **Answer: D**

23. Which of the following occurs during the dark (light independent) reactions of photosynthesis?
A. Splitting of water molecules into hydrogen and oxygen
B. Energy is stored in ATP and $NADPH_2$
C. Synthesis of glucose
D. Absorption of light energy

Explanation: The light independent stage of photosynthesis result in the synthesis of glucose.

$$6CO_2 + 24H^+ \rightarrow C_6H_{12}O_6 + 6H_2O$$

Answer: C

24. During light reaction of photosynthesis, light energy is used primarily to:
A. Chemically combine carbon and oxygen.
B. Chemically combine hydrogen and carbon di oxide.
C. Split water into hydrogen and oxygen.
D. Split maltose in to glucose molecules.

Explanation: During the light dependent stage of photosynthesis, light energy is used to split water molecule into oxygen molecule and hydrogen ion.

$$12H_2O \xrightarrow[chll]{light} 6O_2 + 24H^+$$

Answer: C

25. The two functions of the leaf as a photosynthetic organ are to:
A. Capture light energy and carbondioxide.
B. Capture carbon di oxide and oxygen.
C. Capture light energy and water vapor.
D. Release oxygen and carbondioxide.

Explanation: The leaf as a photosynthetic organ captures light energy and carbon dioxide that are used during synthesis of glucose in photosynthesis.

Answer: A

26. The oxygen released by green plants during photosynthesis comes from the:
A. CO_2 molecule
B. Atmosphere O_2
C. H_2O molecule
D. Chlorophyll

Explanation: The oxygen that is released into the atmosphere from the breakdown of water molecule.

$$12H_2O \xrightarrow[chlorphyll]{light} 6O_2 + 24H^+$$

Answer: C

27. The role of chlorophyll in the process of photosynthesis is:
A. To capture carbondioxide and water
B. To capture carbondioxide

C. To capture sun light
D. To capture water

Explanation: Chlorophyll is a pigmented substance that is found in the chloroplast of the leaf and is used to capture sunlight.

Answer: C

28. Plants give out CO_2 when its dark because:
A. Leaves are respiring
B. Leaves have excess CO_2
C. There is excess CO_2 when it is dark
D. No difference in CO_2 up take whether dark or bright.

Explanation: Plants give out CO_2 when its dark due to the reason that they are respiring. **Answer: A**

29. Oxygen is the bi product of photosynthesis. Experimental results show that it originates from:
A. Water
B. Caron di oxide
C. Carbon hydrate
D. Sun light

Explanation: Oxygen which is the bi-product of photosynthesis originates from water molecules. **Answer: A**

30. Which of the following is true concerning the light independent stage of photosynthesis?
A. Carbon dioxide and hydrogen are combined to form glucose.
B. Water molecules are split to hydrogen and oxygen.
C. Formation of NADPH and ATP.
D. It occurs in the grana of the chloroplast.

Explanation:
During the light independent stage of photosynthesis carbondioxide and hydrogen are combined to form glucose.

Answer: A

31. During which stages of photosynthesis is oxygen produced?
A. Cyclic photo phosphorylation
B. The light dependent reactions
C. Carbon fixation
D. The light independent stage of photosynthesis.

Explanation: Oxygen is produced during the light dependent stage of photosynthesis. **Answer: B**

32. The light dependent photosynthetic reaction produces:
A. Energy, hydrogen and oxygen.
B. Glucose and carbon di oxide

C. Glucose, and carbon dioxide
D. Hydrogen water and carbon di oxide.

Explanation: The light dependent stage of photosynthesis produces:

- ✓ Energy
- ✓ Hydrogen
- ✓ Oxygen

33. What are the major requirements of photosynthesis?

Explanation:

The major requirements or needs of photosynthesize are:

- ✓ Sunlight
- ✓ Chlorphyl
- ✓ Water
- ✓ Carbon di oxide

34. Explain the role of water during the process of photosynthesis?

Explanation: The role of water in photosynthesis is it serves as a source of hydrogen ion found in a glucose molecule.

35. What is the relationship between photosynthesis and respiration?

Explanation: During photosynthesis carbondioxide is utilized and oxygen is released. While, during respiration oxygen is utilized and carbondioxide is released. Thus, the bi – product of photosynthesis is needed during respiration and vice versa.

36. Write the chemical reaction of:

- ✓ Light dependent reaction
- ✓ Light independent reaction
- ✓ Overall reaction of photosynthesis

Explanation:

✓ The chemical reaction during light dependent reaction is:

$$12H_2O \xrightarrow[\text{chlorophyll}]{\text{sun light}} 6O_2 + 24H^+$$

✓ The chemical reaction during light independent reaction is:

$$24H^+ + 6CO_2 \longrightarrow C_6H_{12}O_6 + 6H_2O$$

✓ The overall reaction of photosynthesis is the combination of the above two reactions:

$$6H_2O + 6CO_2 \xrightarrow[\text{chlorophyll}]{\text{sunlight}} C_6H_{12}O_6 + 6O_2$$

37. Which one of the following is NOT the requirement of photosynthesis?

A. Oxygen
B. Carbon dioxide
C. Water
D. Sun light

Explanation: Carbondioxide water, sun light and chlorophyll are the requirements or needs of photosynthesis, while oxygen is the bi-product of photosynthesis. **Answer: A**

38. The carbon and oxygen that are found in a glucose molecule produced by photosynthesis comes from:
A. Chlorophyll
B. Carbon monoxide
C. Water
D. Carbon di oxide

Explanation: Carbon di oxide is a source of carbon and oxygen that is found in a glucose molecule. **Answer: D**

39. How do photosynthesis balance concentration of O_2 and CO_2? It
A. Releases carbondioxide and absorbs oxygen from the atmosphere.
B. Oxidizes and removes carbondioxide from the atmosphere.
C. Converts oxygen ozone and carbondioxide to carbonic acid.
D. Releases oxygen and absorbs carbondioxide from the atmosphere.

Explanation: During the process of photosynthesis organisms absorb carbondioxide from the atmosphere and release oxygen gas to the atmosphere. **Answer: D**

40. What is the role of chlorophyll in photosynthesis. To absorb
A. Light energy
B. Carbondioxide
C. Water molecule
D. Oxygen gas

Explanation: The role of chlorophyll in photosynthesis is to absorb light energy. **Answer: A**

Post Test 2

1. Which of the following organisms do **NOT** carry on photosynthesis?
A. Plants
B. Bacteria
C. Some algae
D. Animals

2. Solar or radiant energy is involved in life because _____.
A. Plants depend upon sunlight to synthesize food by photosynthesis.
B. Plant bodies became fossil fuels.
C. Sunlight heats the earth's surface and provides the physical conditions for life.

D. All of the above are ways in which life is dependent upon the sun.

3. The correct description of the relationship between photosynthesis and the living world is:
   A. Herbivores, carnivores and omnivores are autotrophs since they can "eat for themselves."
   B. Heterotrophs are called "hetero" because they change sunlight energy into chemical energy.
   C. All heterotrophs are ultimately dependent upon food produced by autotrophs.
   D. Herbivores are autotrophs, carnivores are heterotrophs, and omnivores are half autotrophs and half heterotrophs.

4. Plants and some algae carry on photosynthesis within _____.
   A. Chloroplast C. Lysosome
   B. Mitochondria D. Golgi body

5. Thylakoids and grana are found within organelles called_____.
   A. Ribosome C. Golgi body
   B. Chloroplast D. Lysosome

6. The ability of plants, algae, and a few kinds of bacteria to make their own food in the presence of sunlight is called _____
   A. Respiration C. Photosynthesis
   B. Excretion D. Germination

7. Which of the following molecules is produced during the light – dependent reactions of photosynthesis and used in the light-independent reactions?
   A. Oxygen and glucose
   B. Carbon dioxide, NADPH, and ATP
   C. Oxygen, NADPH, and ATP
   D. NADPH and ATP

8. NADPH and ATP are used in the _____ reactions of photosynthesis.
   A. Light-dependent C. Fermentation
   B. Light independent D. None

9. In which photosynthetic reaction is carbondioxide reduced to form a carbonhydrate?
   A. Light independent C. Primary phase
   B. Light dependent D. Glycolysis

10. In which of the following stages of photosynthesis will water split?

A. Light dependent B. Light independent

11. In which of the following stages will oxygen produced during photosynthesis?
A. Light independent C. Dark reaction
B. Light dependent

12. Which molecules transports hydrogen from the light-dependent reactions to the light-independent reactions?
A. FAD B. NAD C. NADP D. Water

13. Where in plants does most photosynthesis occur?
A. Flower C. Stem
B. Leaf D. Root

14. What is the starting source of the carbon in the glucose produced by photosynthesis?
A. Water C. ATP
B. Chlorophyll D. Carbon di oxide

The Need for Photosynthesis

As mentioned in the previous section, the process of photosynthesis requires water, sunlight and carbondioxide. As a result it produces glucose and oxygen molecules.
Under this section we will see experimentally the needs and products of photosynthesis.

Testing a leaf for Starch

The glucose molecule produced by the process of photosynthesis is temporarily stored in the leaf in the form of **starch**.

> Starch is a complex sugar that is made up of a long chain of glucose molecules.

To check whether the leaf is photosynthesizing or not, it must be tested for the presence of starch. The steps used for starch test in a leaf are:

1) Detach a leaf from a plant
2) Put the leaf in a boiled water for 30 seconds to kill the leaf.
3) Place the leaf in **ethanol solution** to remove the chlorophyll pigment.
4) Dip the decolonized leaf in to cold water to soften it.

5) Add the iodine solution on the leaf. If the color of the leaf turns blue black, it shows the presence of starch in the leaf.

The need for light

During the process of photosynthesis light is needed for:

- Splitting water into hydrogen and oxygen

$$12H_2O \xrightarrow[\text{chloroplast}]{\text{sun light}} 24H^+ + 6O_2$$

- To form the chemical energy found in glucose formed by photosynthesis.

To demonstrate sunlight is essential for photosynthesis:

1) Take a healthy plant growing in a pot.
2) Destarch a plant by putting it in a dark place where it cannot get light fro few days. As a result, the leaf loses its starch content. This process is called **destarching**.
3) Test one leaf for starch, in order to make sure that it does not contain starch.
4) Cut a simple shape (for instance 'E' shape) from the aluminum foil for black paper. The cut shape should allow light to pass through it.
5) Fasten the foil or black paper firmly over both sides of a leaf.
6) Place a plant in sunlight for 3 – 4 hours.
7) Detach the leaf and test in it for starch.

Result – only the cut part 'E' which received light turns blue black showing presence of starch. Therefore, it can be concluded that:

- Only the part which received light turns blue black and photosynthesized.

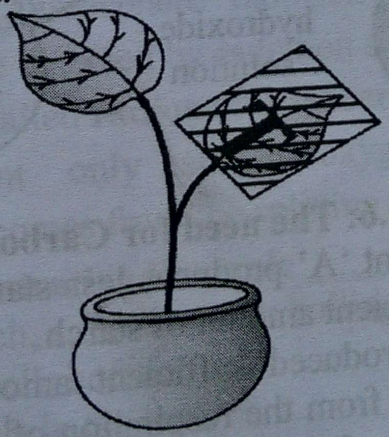


Figure 10.5: Need for Light

The Need for Carbon di oxide

Carbon di oxide is need during photosynthesis to incorporate the carbon and oxygen that is found in glucose molecule.

$$6CO_2 + 24H^+ \rightarrow C_6H_{12}O_6 + 6H_2O$$

To demonstrate that plants need carbondioxide for photosynthesis:

1. Take two plants growing in a pot. Plant 'A' and plant 'B'.
2. Destarch both plants by placing them in a dark place for few days.
3. Put both plants in a bell jar which do not allow air to enter. Place potassium hydroxide solution around plant A.

Potassium hydroxide has an ability to absorb carbon dioxide from surrounding air.

4. Place both plants in light for few hours.
5. Test leaves from both plants for starch.

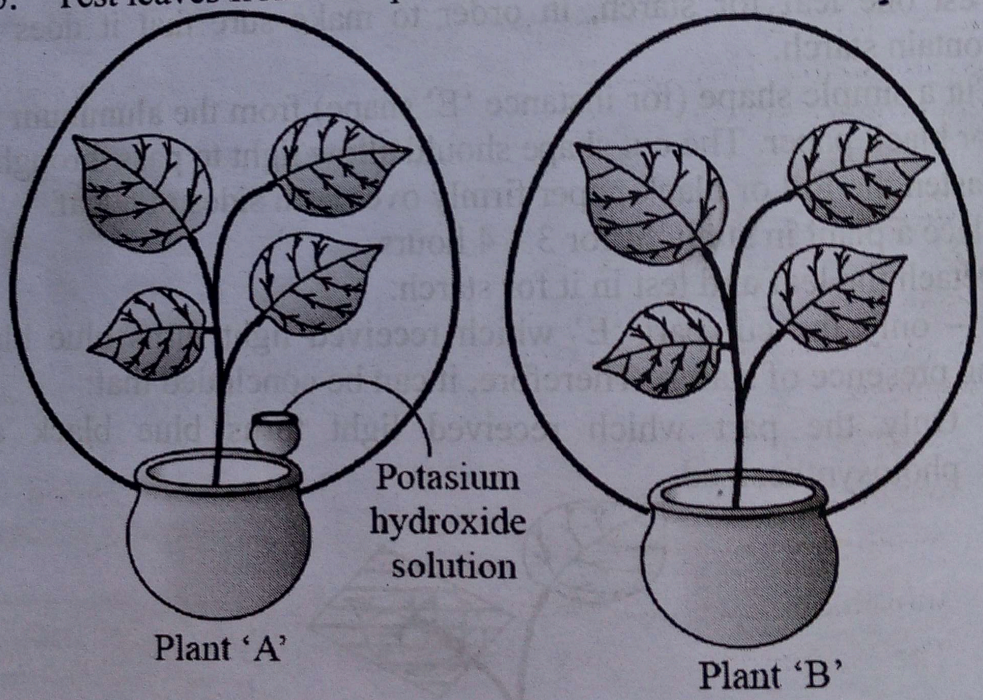


Figure 10.6: The need for Carbon dioxide

Result – the leaf of plant 'A' produces less starch, while the leaf from plant 'B' produces sufficient amount of starch.

Therefore, plant 'B' produced sufficient among of starch by using carbondioxide produced from the respiration of the plant. But plant 'A' produced less amount of starch because the potassium hydroxide absorbed the carbondioxide that made the plant to face the shortage of CO_2.

The Need for Chlorophyll

Chlorophyll is needed during photosynthesis to absorb sunlight which is used as the source of energy. To demonstrate the need to chlorophyll during the process of photosynthesis.

1. Take a plant with variegated leaf.

> **Variegated leaf** is a leaf which is NOT completely green and has yellow or white spots.

2. Destarch a leaf by placing the plant in a dark place for 3 to 4 hours.
3. Place the plant in light only for few hours.
4. Take a leaf and test it for starch.

Result – only the greenish part turns blue black because it is photosynthesizing. However the variegated part has no chlorophyll and it do not produce starch.

The Need for Water

Water is needed during photosynthesis as a source of hydrogen ion that is found in glucose molecule. To demonstrate the need for water during photosynthesis.

- ✓ Supply a plant with a heavy water containing the "O isotope of oxygen. These atmosphere radioactive. The radiation they produced can be detected by radiation sensitive machine. These substances are called **radioactive tracers**.
- ✓ The experiment only shows oxygen produced during photosynthesis comes from the splitting of water molecule.

Illustrative Example

41. In testing for starch, a leaf is put in boiling water, this is to:
    A. Make the cells soft for test.
    B. Kill the cells in the leaf.
    C. Break down starch in to sugar.
    D. Remove the chlorophyll from the leaf.

Explanation: The leaf is put in a boiled water to kill its cells. **Answer: B**

42. Which of the following requirements of photosynthesis can be demonstrated during experiments by using a variegated leaf?
A. Chlorophyll
B. Water
C. Carbon dioxide
D. Light

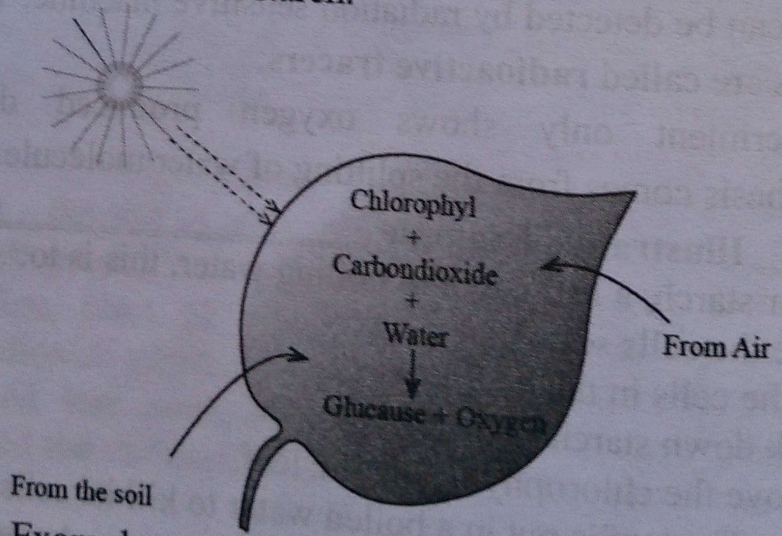
Explanation: Variegated leaf is a leaf which is NOT completely green. It has green and non-green parts. The non – green parts do not have chlorophyll and do not photosynthesize. Thus variegated leaves are used to show the need of chlorophyll in photosynthesis. **Answer: A**

43. It samples of green plants are kept in four different rooms with respect to the color of light, the greatest absorption of light occurs in the room with:
A. Blue light
B. Green light
C. Total darkness
D. Yellow light

Explanation: The greatest absorption by a chlorophyll during photosynthesis occurs at **red** and **blue** spectrum. While, the highest reflection occurs at green light. **Answer: A**

44. If the decolorized leaf did not become blue black when put in iodine solution, the probable explanation would be:
A. The leaf did not have enough starch.
B. The leaf did not contain starch.
C. Both A and B
D. None of the above

Explanation: When iodine is putted on starch it gives a blue black color. Thus, if the decolorized leaf does not turn blue black when iodine is putted on it, this indicates the leaf does not contain starch. **Answer: B**



Example number 5 is based on the above diagram.

45. The above diagram is good representation for the process of :
A. transpiration
B. respiration
C. photosynthesis
D. protein synthesis

Explanation: The diagram shows the process of photosynthesis
✓ sunlight carbon dioxide and sunlight are used in photosynthesis to produce glucose. **Answer: C**

Examples number 46 and 47 are based on the information given below.

A student kept fresh green leaf in a boiling water for about five minutes. He they decolorized it by placing in a warm alchol.

46. The student observed that the absolute alcohol turned greenish in colour. This is because:
A. Starch was extracted from the leaf while boiling.
B. Chlorophyll pigment was extracted from the leaf.
C. The heat applied changed the color of the alcohol.
D. The reaction between starch and absolute alcohol produced a green colored substance.

Explanation: The effect of warm alcohol makes the chlorophyll to be extracted from the leaf. **Answer: B**

47. The decolorized leaf turned blue black color up on the addition of iodine solution. This must be due to:
A. the nature of alchol used
B. the leaf being photosynthetic initially
C. the leaf being non-photosynthetic
D. the leaf not being exposed to light

Explanation: A photosynthetic leaf produces sugar which is stored in the form of starch. Thus, when iodine is added on it after being decolorized, its color turns blue black. **Answer: B**

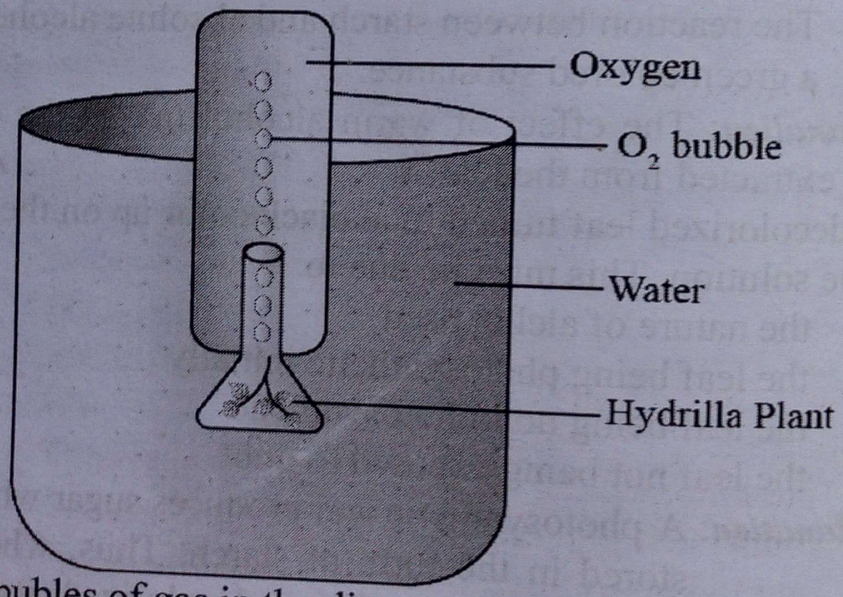
48. Suppose plant 'A' and plant 'B' are destarched. Plant 'A' has no carbon dioxide supply, while plant 'B' has normal level of carbon dioxide. Both are given sunlight for six hours and tested for the presence of starch. The amount of starch produced by plant:
A. 'A' is less than plant 'B'
B. 'B' is less than plant 'B'
C. 'A' is equal to plant 'B'
D. 'A' is greater than or equal to plant 'B'

Explanation: Plant 'B' produces starch and plant 'A' don't. This is due to the reason that plant 'A' has no carbon dioxide supply and it cannot photosynthesize. **Answer: A**

49. Chemical substance that is used to absorb carbon dioxide in the activity to show its importance in photosynthesis is:
    A. Potassium hydroxide solution
    B. Copper sulphate solution
    C. Iodine solution
    D. Benedict's solution

Explanation: Potasium hydroxide solution has an ability to absorb carbon dioxide. Thus, it is used in testing the need for carbon dioxide during photosynthesis. **Answer: A**

Examples number 50 and 51 are based on the following diagram. The diagram shows a branch of elodea placed in a test tube with water. The test tube was then placed in sunlight for six hours.



50. The bubles of gas in the diagram are mainly composed of:
    A. Carbon monoxide
    B. Carbon dioxide
    C. Nitrogen
    D. Oxygen

Explanation: The bubbles of gas are oxygen molecules which produced as a bi-product of photosynthesis.

$$6Co_2 + 6H_2o \xrightarrow[\text{sunlight}]{\text{chlorophyll}} C_6H_{12}o6 + 6o_2$$

Answer: D

51. The bubbles of gas in the diagram are mostly results of:
    A. Photosynthesis
    B. Chemosynthesis
    C. aerobic respiration
    D. anaerobic respiration

Explanation: The bubbles of oxygen gas that are released in to the atmosphere are bi-products of photosynthesis.

Answer: A

52. In the leaves of the elodea plant, the energy from the sun is first used to make?
   A. Lipids
   B. Proteins
   C. Nucleic acid
   D. Carbohydrates

Explanation: In the process of photosynthesis plants primarly make carbohydrate (glucose). Proteins (anion acids) are synthesized from the combination of glucose and mineral ions absorbed from the soil.

Answer: D

Post Test – 3

Part I. Matching

Colum 'A'	**Column 'B'**
1. Destarching	a. a reagent used to test for the presence of starch in a leaf.
2. Iodine solution	b. a pigment that is used to absorb light during photosynthesis.
3. Potassium hydroxide	c. a complex sugar that is stared in a leaf.
4. Chlorophyll	d. a chemical that is used in demonstrating the need for carbon dioxide during photosynthesis
5. Starch	e. removal of starch from a leaf.

Part II. Choose the best option from the given alternatives.

1. In testing a leaf for starch;
   A. put it in a boiled water
   B. immerse it into alcohol
   C. add iodine solution on it
   D. bask it in a sun for ken hours

2. While adding iodine solution on a starch it gives a _____ colour
   A. red
   B. colures
   C. Blue black
   D. white

3. For deutarching a leaf:
   A. Put it in a dark place for few hours.
   B. Bask it in the sun
   C. Put it in alchol
   D. None of the above

4. A varigated leak is used in testing the need for _____ during photosynthesis.
   A. Carbon dioxide
   C. Sunlight

B. Water
D. Chlorophyll

5. A leaf that is Not completely green is:
   A. Carotene
   B. Variegated
   C. Xanthophyll
   D. None of the above

10.2 Transport in Plants

Just like animals plants need the transportation of different materials such as water, minerals, gases, nutrients, etc.

Simple plants such as mosses and liverwort use transport mechanisms such as Osmosis and diffusion to transport substances such as:

- ✓ Carbon dioxide
- ✓ Water
- ✓ Minerals
- ✓ oxygen
- ✓ products of photosynthesis

Simple plants do not have special transport system.

In higher plants:

- ✓ Gases such as carbondioxide and oxygen move by the process of diffusion.
- ✓ Transportion of water, mineral and food takes place through specialized tissues called **vascular bundles (xylem** and **phloem)**

Plants absorb water and minerals from the soil. The process occurs by **Osmosis** through specialized **root hair cells**. The root hair cells have an increased surface area to enable the plant absorb more water by osmosis.

Osmosis is the movement of water molecules from the region of higher concentration of water to a region where there is less concentration of water through a semi- permeable membrane.

Absorption of water by osmosis occurs through root hair cells because?

- ✓ Soil has high water concentration than the root hair cells. Thus, water is moves from the soil to the root hair cells by osmosis.
- ✓ Once water entered the root hair cells, it moves from one cell to another cell by osmosis until it reaches xylem vessels.

Active Transport in Plants

Active transport is the movement of materials from the region of lower concentration to the region of higher concentration, against concentration gradient.

Plants need active transport during:

- ✓ Absorption of mineral ions from the soil.
- ✓ Transportation of organic substances from the leaf to different parts of a plant. This process is known as **translocation**.

The reason why absorption of minerals ions require active transport is:

- ✓ The soil environment is less concentrated (more dilute) than the root hair cells. Thus, minerals move by active transport from a less concentrated area (soil) to a more concentrated area (root hair cells)

Translocation occurs by using active transport system because:

- ✓ Organic substances (food) moves from the leaf where there is relatively less concentration of organic substances to different parts of a plant where there is more.

For active transportation to occur, there is a need of expenditure of energy from a cell.

Double Transport System

The transportation system of plants occurs through a system of vessels known as **vascular bundles**. The vascular bundle consists of two system of vessels; namely:

- ✓ xylem vessel
- ✓ phloem vessel. Thus the transport system of plants is called a **double transport system**.

Xylem Vessel

Xylem vessel is responsible for transporting or carrying water and **mineral ions** from the soil to different parts of a plant. It sometimes known as a **dead vessel.**

The arrangement and structure of xylem and phloem vessels vary between monocot and dicot plants.

- ✓ **In dicot plants**
  - **xylem and phloem vessels appear in rings.**

- Phloem vessels are found just underneath the bark.
- Xylem vessels are found inner to phloem vessels.

✓ In monocot plants:

- Xylem and phloem vessels appear in scattered manner not in ring.

Phloem Vessel

Phloem vessel is involved in the transportation of organic materials made by plants during the process of photosynthesis. They transport organic materials from the leaf to different parts of a plant such as stem, root, fruit, etc.

~~Xylem~~ Phloem vessel is sometimes known as a **living tissue.** The process in which organic materials get transported in a plant is known as **translocation.**

Translocation requires expenditure of energy from a plant. Thus, it is an **active transport system.**

The organic substances which are transported through phloem vessel are needed for making new plant materials and stored in storage organs of a plant such as **root, stem,** etc.

In dicot plants if a complete ring of bark of a tree is nibbled (removed) specially in young plants, the organic substances are prevented from reaching some parts of a plant and this results in a death of a plant.

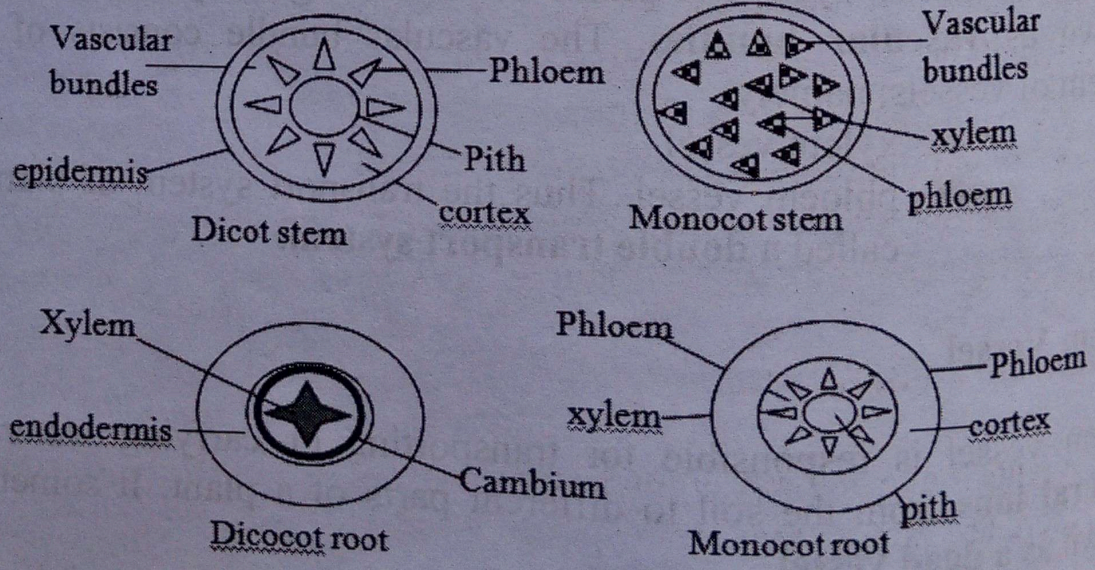


Figure 10.7 Dicot and monocot root and stem (cross sectional area)

The need for transport in plants

Plants need to transport substances such as water, mineral ions organic substances to enable the plant carry out all the metabolic process that is required by a plant as an organism.

i. The need for transportation of water

Plants need water for various purposes. The major reasons are:

- ✓ For carrying out **photosynthesis** water is a source of hydrogen ion (H+) found in a glucose molecule.
- ✓ Plants cells need to be turgid and firm to be in a hypotonic condition which is a conducive situation for them.

ii. The need for minerals ions in plants

Plants need mineral ions mainly for making organic substances such as protein and also other molecules with in the cells.

iii. The need for transportation of organic substances

Plants need glucose for cellular respiration which provides energy for the plant. Thus, transportation of sugar is important in plants.

Plasts also need to store sugar (glucose) in the form of starch which is osmotically inactive (insoluble). Thus, the sugar is transported from the leaf to different parts to be stored.

Plants store starch at different storage organs, for example:

- ✓ root
- ✓ leaves
- ✓ stem

Storage organs are important for a plant to survive difficult conditions. Starch which is stored can be converted in to glucose (sugar)

- when needed by a plant mostly
- when the plant is not photosynthesizing

Plants also store starch in their fruits and seeds, which are sources of energy for humans and other animals.

The transpiration Stream

Some plants have short height, while others are several meters tall. Thus, there are factors that are responsible for the movement of water up along the height of the stem. The major factor responsible for the movement of water up the stem is **transpiration pull**.

Transpiration – is the evaporation of water from a plant which occurs mainly through the leak.
Transpiration pull – is a force that is responsible for moving water up along the length of the stem.

In addition to transpiration pull, there are also other factors that assist the movement of water along xylem vessel in the stem. These are:

- ✓ **Adhesive force (Adhesion)** – is an attraction between water molecules and xylem vessels.
- ✓ **Cohesive forces** – is an attraction between water molecules (polar substances).
- ✓ **Capillarity** – is the ability of water to rise in fine tubes.
- ✓ **Root pressure** – is the pressure created in the root as a result of turgidity of the root cells when water enters in to them from the soil.

Factors affecting the rate of transpiration

Transpiration causes a plant to evaporate large amount of water. This exeedes the water that is incorporated in the use by plants.
There are number of factors that largely affect the rate at which transpiration takes place. These are;

- ✓ **Temperature** – as the temperature increase, the rate of evaporation of water from the leak (transpiration) also increases.
- ✓ **Light** – makes the stomata to stay open. This is due to the reason that photosynthesis occurs in the presence of light. Thus, stomata opens for exchange of gases for photosynthetic activity. This causes water to evaporate by transpiration.
- ✓ **Number of stomata** – as the number of stomatal opening in the leaf increases, the rate of transpiration also increases.
- ✓ **Air movement (wind)** – water around the stomata opening is swept away from the leaf by air movements or wind which causes the water to diffuse out.
- ✓ **Humidity** – when the concentration of water vapour in the atmosphere is saturated, less amount of water diffuses (evaporates) in to the atmosphere from the leaf.

Adaptation of plants to reduce water loss

Plants lose most of the water which they absorb from the soil by evaporation. Evidences show that about 99% of water that is absorbed

from the soil is lost by transpiration. Thus, plants need to reduce the rate of transpiration through various adaptations. These include:

i. **Waxy cuticles** – water proof structures found on the surfaces of the leaf. They are thick and shiny for plants that grow in hot climate.
ii. **Reduced number of stomata**
iii. **Stomata on the underside of the leaf** – large number of stomata is found on the underside of the leaf than the upper surface/ layer /of the leaf. This reduces excessive loss of water by transpiration.
iv. **Reduced number of leaves** – this reduces the probability of transpiration to occur.

Transpiration and agriculture

In order to save the water requirement, specially in areas where there is limited water resource, Farmers must work or device towards the mechanisms that reduce water loss. Therefore transpiration can be reduced by

- growing plants which transpire less.
- reducing air movements over the crop by planting taller plants in rows along the direction of wind break.
- destroying unwanted weeds.
- removing dying or damaged leaves.

Illustrative Example

53. Which of the following factors reduce the rate of transpiration?
A. High temperature C. Light
B. Higher humidity D. Wind

Explanation: The rate of transpiration can be reduced by:
- ✓ low temperature
- ✓ higher humidity
- ✓ less light intensity
- ✓ steady air (less windy)
- ✓ reduced number of stomata.

Answer: B

54. In plants, water rises beyond the point supported by the atmospheric pressure mostly because of:
A. capillarity C. diffusion
B. gravity D. active transport

Explanation: Capillarity is the ability of water to rise in fine tubes.

55. Up take of mineral ions from the soil in to the plant is by:
A. active transport C. translocation
B. osmosis D. passive transport

Explanation: plants takes in minerals from the soil by active transport which requires expenditure of energy. This is due to the reason that; soil is less concerntrated than the root's concentration. Accordingly, minerals move against concentration gradient. **Answer: A**

56. The rate of transpiration is high:
    A. when the rate of evaporation is low
    B. in cold and humid condition
    C. in rainy season
    D. in dry and windy condition

Explanation: The rate of transpiration is high when:
- ✓ the rate of evaporation is high
- ✓ in warm and windy condition.
- ✓ in dry season.

Answer: D

57. The rate of transpiration is inversely proportional to:
    A. Temperature C. Humidity
    B. Light intensity D. air movement

Explanation: The rate of transpiration is inversely proportional to humidity. When the rate of transpiration increases, humidity decreases. **Answer: C**

58. What do xylem vessels carry?
    A. Water and mineral salts up stems from the roots.
    B. Water and mineral salt down stems to the roots.
    C. Dissolved sugars up stems from the roots.
    D. Dissolved sugars down stems to roots.

Explanation: Xylem vessels are used to carry or transport water and mineral ions up the stem from the root. **Answer: A**

59. Which of the following structure is responsible for absorbing water from the soil?
    A. Phloem C. Xylem
    B. Cortex D. root hair

Explanation: Water is absorbed from the soil by structures on the root known as **root hairs.** **Answer: D**

60. Girdling of a tree by completely removing a ring of bark most directly affects the process of:
    A. Translocation C. Photosynthesis
    B. Transpiration D. Cellular respiration

Explanation: Gridling of a tree bark causes the removal of phloem tissue. This affects the transportation of food (organics) substances which is called **translocation.**

61. The main factor responsible for the movement of water from the roots to the leaf parts of the tallest tree is the **Answer: A**
   A. high pressure near the roots
   B. strength of xylem cells
   C. transpiration pull
   D. concentration of water in the soil.

Explanation: There are several factors that are responsible for the movement of water along xylem vessel. However the main factor is **transpiration pull.**

62. Plants transpire through **Answer: C**
   A. the lower surfaces of the leaf only
   B. the upper surface of the leaves only
   C. the lower or upper or both surfaces of the leaves
   D. none of the above

Explanation: Plants mainly transpire through the lower surface of the leaf. This is because the largest number of stomata opening through which transpiration takes place is the lower surface of the leaf. **Answer: A**

63. The inner most part of a root is occupied by:
   A. Cambium
   B. Phloem cells
   C. Pith cells
   D. Xylem cells
   E. Endodermal cells

Answer: C

64. What is capillarity? It is the:
   A. Osmotic uptake of water in to roots from soil.
   B. Force of attraction between water molecules and the surface of xylem
   C. Rising up of water molecule in a narrow tube.
   D. Force of attraction between water in narrow fine tubes.

Answer: C

65. Transpiration in plants is high in:
   A. Windy condition
   B. Wet condition
   C. Humid condition
   D. Cold condition

Explanation: Transpiration is high in:
   ✓ Windy condition
   ✓ Warmer and high temperature
   ✓ Dry condition

Answer: A

66. What is the impact of transpiration in agricultural plants?
A. higher transpiration rate increases the chance of growth of agricultural plants.
B. higher transpiration rate increases the chance of agricultural plant wilting.
C. Very low transpiration increases growth of agricultural plants.
D. Higher transpiration increases the activities of agricultural plants cells.

Explanation: Higher rate of transpiration may cause shortage of water, which may lead the plant to wilting. **Answer: B**

67. Water up take by roots is facilitated by:
A. active transport of water molecules
B. high concentration of water in the roots
C. the large vacuoles in the roots and leaves
D. high surface area of root hairs

Explanation: Root hairs have increased surface area to absorb more water by the process of osmosis. **Answer: D**

68. List out the forces that assist the transpiration pull.

Explanation: Transpiration pull is the main factor responsible for the movement of water along the stem through xylem vessel. It is assisted by forces such as:

- ✓ Adhesion
- ✓ Cohesion
- ✓ Capilarity
- ✓ Root pressure

69. Define the following:

- ✓ Adhesion
- ✓ Transpiration
- ✓ Translocation
- ✓ Capilarity

Explanation:

- ✓ **Adhesion** – is the attraction force between water molecule and xylem vessel.
- ✓ **Transpiration** – is the evaporation of water from a plant mainly through the leaf.
- ✓ **Translocation** – is the movement of food/ organic substance/ from the leaf to different parts of a plant.
- ✓ **Capillarity – is the rise of water in narrow fine tube.**

70. **How do plants absorb mineral ions?**
A. By the action of enzymes as carriers
B. Without using the action of enzymes
C. By using energy from the plant
D. Without using energy from the plant

Explanation: plant roots absorb mineral by the process of active transport which requires expenditure of energy. **Answer: C**

71. Which of the following transport processes take place in the phloem? The movement of:
   A. organic substances from the leaves to other parts
   B. water from soil to the roots
   C. minerals from the roots to the leaves
   D. Co_2 from the air to the palisade mesophyll

Explanation: Phloem tissues are used to transport organic substance /food/ from the leaf to different parts of the plants. The process is called **translocation**. **Answer: A**

Post test 4

Part I. Choose the best answer from the given alternatives.

1. Transportation of substances such as water and minerals in plants take place through:
   A. Root hair
   B. Xylem vessel
   C. Phloem vessel
   D. Capillarity tube
2. Absorption of water by root hair cells from the soil occurs by the process of:
   A. Osmosis
   B. Active transport
   C. Diffusion
   D. Passive transport
3. In plants the largest amount of the water that is absorbed from the soil is evaporated through the leaf. The process is called:
   A. Transformation
   B. Translocation
   C. Transpiration
   D. Absorption
4. The role of phloem in plants is:
   A. To transport water from the root to different parts of the plant
   B. To transport starch from the leaf to parts of a plant
   C. To carry minerals along different parts of a plant
   D. To remove excessive organic substances from a plant
5. The absorption of minerals ions from the soil occurs:
   A. Along concentration gradient
   B. Against concentration gradient
   C. Requires expenditure of energy
   D. B and C are correct

6. The movement of organic substances from the leaf to different parts of a plant is:
   A. Translocation
   B. Transpiration
   C. Capillarity
   D. Root pressure
7. Which of the following is a passive transport?
   A. Absorption of water by root hair from the soil
   B. Absorption of minerals by the root from the soil
   C. Translocation
   D. All are possible answers
8. The rise of water in fine tube is known as:
   A. Adhesion
   B. Capillarity
   C. Cohesion
   D. Transpiration
9. Which of the following is not responsible for the movement of water along the height of the stem?
   A. Capillarity
   B. Root pressure
   C. Adhesion
   D. Translocation
10. The attraction between water molecules is:
    A. Adhesion
    B. Capillarity
    C. Cohesion
    D. Transpiration
11. The major force responsible for the movement of water along the stem though xylem vessel is:
    A. Cohesion
    B. Transpiration pull
    C. Adhesion
    D. Capillarity

10.3 Response in Plants

Plants respond to changes that occur in their environment.

Stimulus – environmental change that initiates an organism to respond.
Response – the reaction of an organism to the stimulus.

Co- ordination and response in plants is managed by **hormones**. Plants hormones are sometimes called **phytohormones.**
Phytohormones have various effects on plants. These include:

- ✓ Co ordination
- ✓ Cell division
- ✓ Cell elongation
- ✓ Fruit ripening
- ✓ Growth

Phytohormones

The major phytohormones are:

- ✓ Auxin
- ✓ Gibberellic acid
- ✓ Cytokinin
- ✓ Ethylene
- ✓ Abscisic acid

i. Auxin

The best known auxin is **Indol Acetic Acid (IAA)**. It is involved in growth of plants in various ways. These include:

- ✓ Stimulates elongation of new plant cells.
- ✓ Induce formation of roots on cuttings.
- ✓ Seed germination
- ✓ Allows the plant to grow taller, which increases exposure to sun
- ✓ Prevents falling off fruit before ripening

ii. Gibbereellins

Gibberellin also affects the growth of plants in different ways. These are:

- ✓ It is stimulates the growth of stems.
- ✓ It helps seeds to break their dormant period and start to germinate by stimulating enzymes used to break down a starch stored in a seed
- ✓ It makes dwarf plants to grow in to normal size.

iii. Cytokining

Cytokinin plays role in the growth and coordination of plants by:

- ✓ Stimulating the cell division.
- ✓ Cause differentiation of roots, stems, leaves and buds of a plant together with the auxin.
- ✓ Involved in breaking seed dormancy.

iv. Ethylene

Ethylene is a gaseous hormone which is involved in:

- ✓ Ripening of fruit.
- ✓ Prevention of falling of leaves and fruits in some plants.

v. Absicsic acid (ABA)

ABA is important in:

- ✓ Inhibiting the growth of plants.
- ✓ Playing role in leaf fall.
- ✓ Involved in seed dormancy and may involved in geotropism.

Apical dominance and pruning

Apical dominance is a condition where that growth of the apical part of a plant dominates the growth of lateral branches. It makes a plant long and thin, with only few branches.

Apical dominance makes a plant:

- ✓ incapable of holding heavy loads.
- ✓ affects the quality and quantity of fruits and seed.
- ✓ makes the collection of seeds and fruits difficult.

Pruning is the process of removing the apical dominance by cutting the apical part of a plant. It makes a plant:

- ✓ to produce increased yeald in the quality and quantity of seeds and fruits.
- ✓ Makes the collection of seeds and fruits easy.

Tropic responses

The directional growth movement of plants in response to stimuli is called **tropism**. It involves

- ✓ **Positive tropism** – growth movements of plants towards the direction of stimuli.
- ✓ **Negative tropism** – growth movement away from the direction of the stimuli.

Some examples of tropic responses are:

- ✓ **Phototropism** – is a directional growth movement of plants in response to **light**.
- ✓ **Geotropism** – is a directional growth movement in response to **gravity**
- ✓ **Chemotropism** – is a directional growth movement in response to **chemicals**.
- ✓ **Hydrotropism** – is a directional growth movement in response to **water**.
- ✓ **Tigmotropism** – growth in response to **touch**.

To further clarity let us look at the following example:

i. Plant roots grow down the soil and shoots grow up th ground. There fure:

- The growth of roots down the soil is an example of **positive geotropism** and **negative phototropism**.
- The growth of shoot is a example of **positive phototropism** and **negative geotropism**.

The growth response of plants to gravity can be detected by an apparatus called **klinostaf**.

ii. Growth of root towards damper soil is an example of **positive hydrotropism.**

Tropic responses can be summarized as

Tropism	Stimulus
Phototropism	Light
Geotropism	Gravity
Tigmotropism	Touch
Hydrotropism	Water
Chemotropism	Chemicals

Nastic movements

Nastic movements are non-directional growth of plants in response to stimuli.

Examples of nastic movements include:

- Opening of flowers.
- Falling of fruit and seeds.

Illustrative Example

72. Describe some examples for
    - Positive photropism
    - Positive geotropism
    - Negative phototropism
    - Positive hydrotropism

Explanation:

	Example
Positive phototropism	Growth of stem towards the sun
Positive geotropism	Growth of the root towards gravity
Negative phototropism	Growth of the root away from the sun
Positive hydrotropism	Growth of the root towards damper soil.

73. Ripening of fruits such as banana is speeded by:
    A. Ethylene
    B. Abscisic acid
    C. gibberellin
    D. indole acetic acid

Explanation: Ethylene is a plant hormone that is involved in:
- ✓ Stimulating the ripening of fruit.
- ✓ Preventing falling of leaves and fruits in plants

Answer: A

74. Which of the following growth substances is synthesized by a plant?
A. Indole acetinate
B. Indole pyruvica acid
C. Indole acetic acid
D. Indole butyric acid
Explanation: Indole acetic acid (IAA) is a main type of auxin which is a plant hormone **Answer: C**

75. Which of the following plant hormone is correctly paired with its function?
A. Ethylene – inhibits fruit ripening
B. Abscisic acid – inhibits fruit ripening
C. Gibberellic acid – inhibits growth
D. Cytokinin – stimulates cell division **Answer: D**

76. Which of the following plant hormones induces roots formation in stem cuttings?
A. Ethylene
B. Cytokinin
C. Gibberellin
D. Auxin
Explanation: Auxin is a plant hormone that is involved in the following activities.
- ✓ Stimulates elongation of new plant cells.
- ✓ Induce formation of roots on cuttings.
- ✓ Seed germination
- ✓ Allows the plant to grow taller.
- ✓ Prevents falling of fruit before ripening **Answer: D**

77. Which of the following is correctly matched with its function?
A. Auxin – responsible for apical dominance
B. Abscisic acid – promotes root formation in cuttings
C. Cytokinins – delays senescence
D. Gibberellins – promotes shoot elongation
Explanation: Apical dominance is caused by the high concentration of auxin on the tip of the shoot of a plant.
Answer: A

78. Plant roots grow towards the sources of water in the soil. This is known as:
A. Chemotropism
B. Geotropism
C. Phototropism
D. Hydrotropism
Explanation: Hydrotropism is the growth response of the plants in response to water. **Answer: D**

79. Some plants need low temperature to flower. This condition is known as:
A. phototropism
B. tropism
C. Senescence
D. Vernalization
Explanation: Vernalization is a low temperature requirement by plants for flowering. **Answer: D**

80. Unlike tropism, nastic movements are in response to:
A. darkness
B. wind
C. non directional stimuli
D. directional stimuli
Explanation: Nastic movement is non – directional growth movement in plants. **Answer: C**

81. When a tendril comes in to contact with an object, it causes growth at the base of the tendril. This response is called:
A. Geotropism
B. Thigmotropism
C. Nastic movement
D. Chemotropism
Explanation: Thigmotropism is a growth in response to touch. Thus the above example shows the growth movement of tendril in response to touch of an object. **Answer: B**

82. Which of the following is an example of hydrotropism?
A. Pollen tubes grow towards ovaries
B. Shoots grow towards light source
C. Roots grow towards a source of water
D. Tendrils grow around stems **Answer: C**

83. Which of the following is NOT true about pruning?
A. It encourages the development of the plants
B. It decreases the size and quality of plants
C. It serves to keep the beuty of the plant.
D. It makes the plant remain in a desired height.
Explanation: Pruning is the removal of the apical dominance. It
✓ encourages the development of plants.
✓ increases the size and quality of plants and their products such as fruits and seeds.
✓ serves to keep the beauty of the plant.
✓ makes the plant to have desired height **Answer: B**

4. Which of the following parts of a plant is characterized by active cell division?
A. Meristem
B. Root cap
C. Phloem
D. Xylem
Explanation: Meristernatic tissue is a part of a plant where rapid cell division and growth take place. **Answer: A**

85. What is the effect of removing apical dominance? It
A. increases stem length
B. causes side shoot growth.
C. prevents fruit formation
D. increases water loss

Explanation: The removal of apical dominance:
- ✓ increases the quality and quantity of fruits and seeds.
- ✓ makes the growth of lateral and apical parts of a plant balanced

Answer: B

86. The tendency of a plant to grow towards water is:
A. Phototropism
B. Hydrotropism
C. Geotropism
D. Chemotropism

Tropism movement	Stimulus
Phototropism	Light
Hydrotropism	Water
Geotropism	Gravity
chemotropism	Chemicals

Answer: B

Post Test 5

Part I. Choose the best answer from the given alternatives.

1. One of the following is NOT the role of auxin:
A. Induce formation of roots
B. Prevents falling of fruit before ripening
C. Stimulates cell division
D. Stimulates elongation of a plant

2. Indole acetic acid is an example of:
A. Gibberellin
B. Auxin
C. Cytokinine
D. Ethylene

3. The phytohormone that makes the plant to grow in to normal size are:
A. Gibberellic acid
B. Ethylene
C. Auxin
D. Abscisis acid

4. A plant hormone that is involved in seed dormancy and inhibition
A. Ethylene
B. Abscisic acid
C. Gibberellic acid
D. Auxin

5. Cell division is stimulated by which plant hormone?
A. Cytolinin
B. Cytokinin acid
C. Ehtylene
D. Gibberelic acid

6. Dominance of the growth of apical parts of the plant over the lateral branches of a plant is:
   A. Lateral dominance
   B. Apical dominance
   C. Dwarfism
   D. Senescence

7. The removal of apical dominance is :
   A. Pruning
   B. Grafting
   C. Cutting
   D. Layering

8. The growth response of plants to gravity is detected by an apparatus called:
   A. Klinostat
   B. Photolight
   C. GPS
   D. Tedolight

II. Matching

Column A	Column B
1 Auxin	A. Ripening of fruit
2 Phototropism	B. Growth response in response to touch
3 Geotropism	C. Directional growth movement in response to light
4 Absicisic acid	D. Stimulates growth of stem
5 Cytokinin	E. Breaking of seed dormancy
6 Gibberellin	F. Inhibits growth
7 Hydrotropism	G. Growth movement in response to water
8 Ethylene	H. Seed germination
9 Tigmotropism	I. Directional growth movement in response to gravity

Germination of seeds

Germination is the process by which a seed changes in to a seeding. For a seed to germinate the following are necessary conditions. These are:

- ✓ **Oxygen** – is needed for a seed to respire inorder to produce energy for germination.
- ✓ **Moisture (water)** – is used to hydrolyze insoluble substances (starch) stored in a seed.
- ✓ **Optimum temperature** – is important for the action of enzymes that assist the process of germination. These include:
  - ➢ **Diastase** – converts starch in to sugars

- **Lipase** – converts stored fats in a seed in to fatty acid and glycerol.

A seed has the following major parts:

- ✓ **Endo sperm** – is a food storage tissue of a seed.
- ✓ **Embryo** – is a part of a seed which develops into a plant. it is sometimes called a young plant inside a seed. It has the following parts;
  - **Plumule** (embryonic shoot) – a part of seed which later develops in to the shoot system.
  - **Radicle** (embryonic root) – is a part which develops into a root system.
  - **Cotyledon** (seed leaves) – serves for storage of starch mainly in dicot seeds.

Based on the number of seeds which they contain seeds can be classified as:

- ✓ **Monocot** are seeds that have one cotyledons.
- ✓ **Dicot** are seeds that have two cotyledons.

Table: Comparision between monocot seed and dicot seed

Monocot	Dicot seed
Has one cotyledon	has two cotyledons
The main food storage is endosperm	Cotyledon and endosperm serve as food storage organs
The embryo remains a very small part of a seed	The embryo takes most of the seed parts.

If a seed donot get necessary conditions for germination it stays dormant. Seeds are sometimes kept dormant in seed banks to preserve them.

Types of germination

There are two major types of germination. These are

- ✓ Epigeal (dicot) germination
- ✓ Hypogeal (monocot) germination

i. Epigeal (dicot) germination

Epigeal germination is a germination in which the cotyledons appear above the soil.

In epigeal germination:

- ✓ **hypocotyl** carries cotyledons and plumule above the soil surface

Hypocotyl is the first leaf like structure that appears on a germinating seed.

Epigeal germination occurs in dicot plants.
Examples: bean, groundaut, cotton, etc.

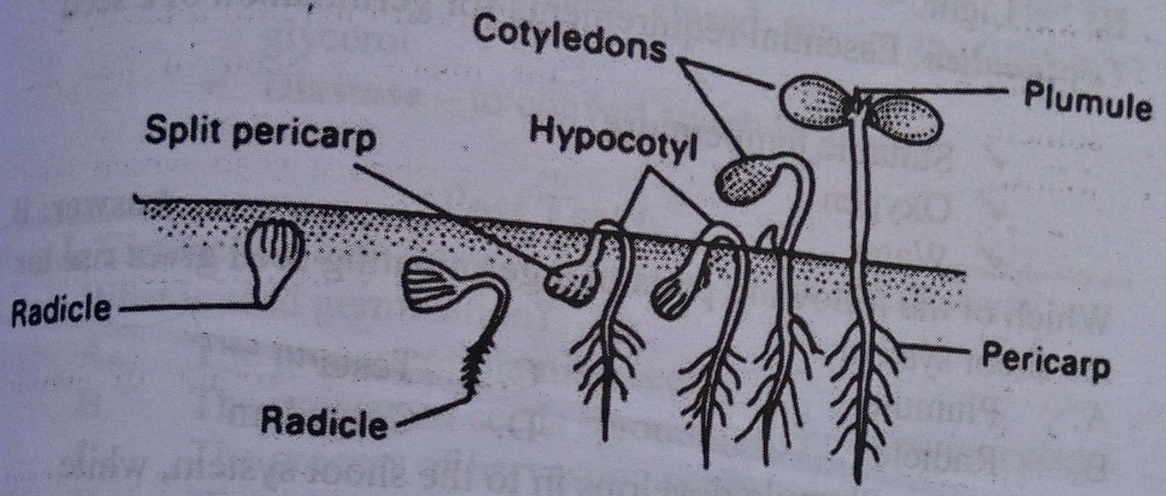


Figure 10.8: Epigeal germination

ii. Hypogeal (monocot) germination

Hypogeal germination is a condition where the cotyledon remains in the soil.

During hypogeal germination:

- ✓ the plumule pushes out of the soil by leaving the cotyledons in the ground.

Hypogeal germination is exhibited by monocot plants
Examples: Maize, wheat, sorghum, etc.

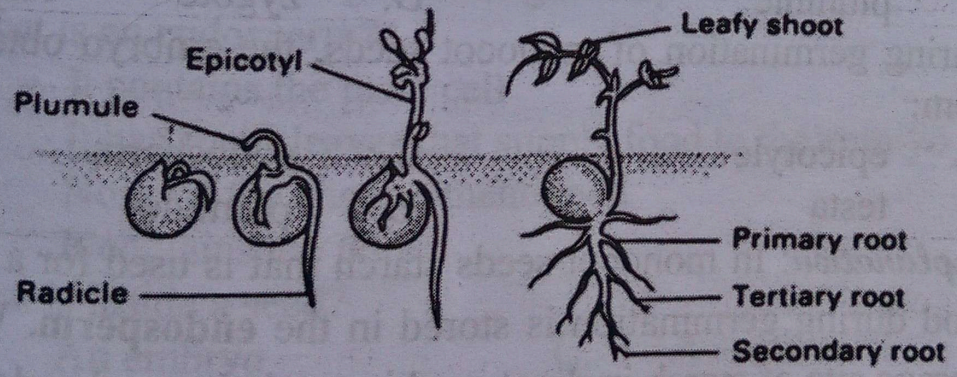


Figure 10.9: Hypogeal germination

Illustrative Example

87. Which of the following is NOT essential for germination of seed?
A. Suitable temperature
B. Light
C. Oxygen
D. Water

Explanation: Essential requirements for germination of a seed are:
- ✓ Suitable temperature
- ✓ Oxygen
- ✓ Water

Answer: B

88. Which of the following parts of a germinating seed gives rise to the shoot system
A. Plumule
B. Radicle
C. Testa
D. Cotyedom

Explanation: Plumule develops in to the shoot system, while radicle develops in to the root system **Answer: B**

89. Which of the following is correct about germinating seed?
A. Cotyledons begin photosynthesis
B. Cotyledons become a shoot
C. Plumule develops in to a root
D. Embryo develop in to a seedling

Explanation: Embryo of a seed becomes a seedling which develops in to a plant. **Answer: D**

90. In germination, the embryo of a seed develops in to:
A. radicle
B. plumule
C. seedling
D. zygote **Answer: C**

91. During germination of monocot seeds, the embryo obtains food from:
A. epicotyle
B. testa
C. endosperm
D. plumule

Explanation: In monocot seeds starch that is used for a source of food during germination is stored in the **endosperm**. While, the storage site of starch in dicot seed is cotyledon and endosperm.

Answer: C

92. List out the major requirements for seed germination

Explanation: The main requirements for seed germination are:
- ✓ Oxygen
- ✓ Moisture
- ✓ Optimum temperature

93. An enzyme that converts starch in to sugar during germination is:
A. Lipase
B. Maltase
C. Amulase
D. Diastase

Explanation: During germination of a seed enzymes that are:
- ✓ **Lipase** – converts stored fat in to fatty acid and glycerol
- ✓ **Diastase** – to convert starch in to sugar.

Post Test 6

Part I. Choose the best answer from the given alternatives

1. What is seed germination?
A. The process of planting seeds
B. The process of seeds sprouting from seeds into plants
C. The process of harvesting seeds
D. The process of cleaning seeds out from fruits

2. What are three of the main parts to a seed?
A. Seed coat, germination, leaves
B. Leaves, stem, cotyledon
C. Seed coat, cotyledon, embryo
D. Cotyledon, embryo, soil

3. What do seeds need to germinated?
A. moisture, oxygen and low temperature
B. moisture, oxygen and the right temperature
C. moisture, oxygen and high temperature
D. moisture, nitrogen and right temperature

4. What is an endosperm?
A. It contains the male cell
B. It has small leaves that supply food to the embryo
C. None of the choices mentioned
D. It contains the pollen

5. What is inside a seed?
A. An embryo
B. Food
C. A small seed
D. Nutrients

6. In most species of plants, food is stored in
A. Radicle
B. Cotyledon
C. plumule
D. zygote

7. The miniature plants contains one or two seed leaves called
A. radicle
B. plumule
C. cotyledon
D. zygote

8. After the root has grown downwards, a small shoot grows upward, towards light. This shoot is known as
   A. radicle
   B. plumule
   C. stem
   D. cotyledon
9. The types of germination on the basis of epicotyls and hypocotyls elongation are
   A. epigeal germination
   B. hypogeal germination
   C. epicotyls germination
   D. both a and b
10. Through seed germination embryo develops into
   A. Complete plant
   B. Zygote
   C. Flower
   D. Seedling
11. The external conditions on which the germination of seeds is dependent are
   A. sufficient food storage and water
   B. live embryo
   C. sufficient food storage
   D. oxygen and water
12. A dicot seed has two cotyledons which:
   A. Prepare food
   B. Store food
   C. digest food
   D. none
13. The outer tough and hard covering of seed is called:
   A. testa
   B. endosperm
   C. radical
   D. cotyledon
14. Inside seed there is a tiny plant is called:
   A. epicotyle
   B. endosperm
   C. embryo
   D. cotyledons

Answer to Post Test – I

1.	C	2.	A	3.	B	4.	A	5.	C
6.	B	7.	B	8.	C	9.	D	10.	D

11. Upper epidermis → palisade mesophyll → mesiphyll → lower epidermis
12. a = midrib, b = venuole, c = vein, d = petiole, e = blade, f = midris, g = apex
13. c 14. e 15. a 16. d 17. b

Answer to Post Test – 2

1.	D	2.	A	3.	C	4.	A	5.	B.
6.	C	7.	D	8.	B	9.	A	10.	A
11.	B	12.	C	13.	B	14.	D		

Answers post Test – 3

Part I.

1. E 2. A 3. D 4. B 5. C

Part II.

1. C 2. C 3. A 4. C 5. B

Answers to post Test 4

1. B 2. A 3. C 4. B 5. D 6. A
7. A 8. B 9. D 10. C 11. B

Answers to post Test 5

Part I

1. C 2. B 3. A 4. B 5. A 6. B 7. A 8. A

Part II

1. H 2. C 3. I 4. F 5. E 6. D 7. G 8. A 9. B

Answers to post Test 6

1. B 2. C 3. B 4. B 5. A
6. B 7. C 8. B 9. D 10. D
11. D 12. B 13. A 14. C

Unit – 11

11. Conservation of Natural Resources

11.1 What is Natural Resource?

Natural resources are resources that exist in nature and are useful for all forms of life including human beings. These resources benefit human beings and other forms of life in different ways. These are:

- ✓ Sources of food.
- ✓ Serves as shelter.

Natural resources can be classified in to two major categories. These are:

- ✓ ***Renewable Natural Resources***- are resources that are capable of being produced indefinitely. They:
  - o Include mostly living things and their products.
  - o Can be reused and replaced easily.

  Examples: animals, vegetation, air
- ✓ ***Non-renewable natural resources***- are resources that cannot be easily made or produced if they are used once.
  - o They include mostly non-living things and cannot be easily replaced.

  Examples: gold, petroleum, coal, diamond, etc.

11.2Conservation and Biodiversity

Conservation- is the wise use of natural resources which involves protection and preservation of the natural environment.
Natural resources can only last if they are used properly in a manageable manner.
If natural resources are not properly utilized and managed their number declines gradually and finally become extinct.

Extinction is the loss of species every where on the world or a particular habitat.

Biodiversity is a measure of wealth of species in a given place or habitat. This includes all types of living things organisms ranging from small microorganisms to big trees and largest animals.
Biodiversity is mostly measured by the number of species breeding at a particular area.

Importance of Biodiversity

Biodiversity is important for all forms of life and the whole ecosystem. Some of the importance of biodiversity are:

- ✓ The balance of nature in an ecosystem. This is related to interaction between organisms and their environment.
- ✓ To develop genetic diversity of crops and animals in order to have an organism that fits with different conditions. It also helps to develop organisms that act as sources of medicine, clothing, food and other useful things.
- ✓ To have areas with admirable and wonderful biodiversity.

Human Impact on Biodiversity

People destroy the biodiversity for several reasons. These are:

- ✓ To grow crops for agricultural purposes.
- ✓ For growing industrial raw materials in a large scale. For example, sugar cane is grown on a huge area as a source of material for sugar factories. In such a case the forest habits is deforested and replaced only by a single species of plant. Such an activity is called **monoculture.** Monocultures:
  - o Eliminates the biodiversity.
  - o Are not attractive to look at.
  - o Do not support wide range of animals and plants.
  - o Increase the risks of disease to spread rapidly.
- ✓ Construction of roads and houses.
- ✓ Release pollutants such as sulfur dioxide that cause acid rain.
- ✓ Hunting wild animals for food and other consumptions.

Therefore conservation of biodiversity or natural resources is useful for the sustainability or the continuity of life on the earth. Thus it is better to establish ways of protecting and preserving the biodiversity.
Some of the ways to conserve biodiversity are:

- ✓ Taking legal actions which protect human beings from killing or harming wildlife and vegetation.
- ✓ Reducing pollutions by reducing the levels of carbon and developing other mechanisms that reduce climatic change.
- ✓ Reducing loss of habitats through deforestation and other human activities.

Illustrative Examples

1. One major current global phenomena is the biodiversity issue. Indicate the statement which is NOT in line with biodiversity concept.
   A. It is concrent about the totality of variability in life and living systems.
   B. It involves the genetic multiplicity within and among species.
   C. It carries the potential resources for food, medicine and industrial materials available to mankind.
   D. Reduction of the existing biological diversity.

   Explanation: Biodiversity
   - is concerned about the totality of variability in life and living system.
   - involves genetic multiplicity within and among species.
   - Carries the potential resources for food, medicine and industrial materials, etc.

   Answer: D

2. Examples of Non-renewable natural resources include:
   A. Solar energy and water
   B. Water and wind
   C. Natural gas and petroleum
   D. Wind and natural gas

   Explanation: Non-renewable natural resources are resources that exist in a limited amount and cannot be easily replaced. Examples are:
   - Natural gas
   - Petroleum
   - Coal
   - Gold
   - diamond, etc.

   Answer: C

3. Which of the following natural resources is different from others?
   A. Natural gas B. Coal C. Gold D. Wood

   Explanation: Natural gas, coal and gold are nor-renewable natural resources, while, wood is a plant which is renewable resource.

   Answer: D

4. All but one is untrue about renewable natural resources:
   A. Exist in unlimited amount
   B. Can be easily replaced
   C. Examples are gold, petroleum, coal.
   D. Are capable of being produced indefinitely.

Explanation: Gold, petroleum and coal are not renewable natural resources. Instead they are non – renewable natural resoruces.

Answer: C

5. Which of the following is an important activity for increasing biodiversity?
   A. Burning of fossil fuel
   B. Cutting of tree
   C. Replanting of tree
   D. Deforestation

Explanation: Biodiversity is a wealth of specis in a given ecosystem. Thus, the following are important activities to enhance biodiversity. These
   - Planting and replanting trees.

Answer: C

6. Biodiversity refers to:
   A. All the natural resources we have on the planet earth.
   B. The variety of living things on the planet earth.
   C. Sustainable use and protection of natural resources.
   D. Identification and scientific description of species of living things.

Explanation: Biodiversity refers to the variety of living things or species on the earth. It includes animals, plants and microorganisms.

Answer: B

7. Why is it necessary to conserve natural resources?
   A. To keep balance in an ecosystem.
   B. To make human beings control all organism.
   C. To enhance the urban development.
   D. To increase mono culturing.

Explanation: Conservation of natural resources is used to keep the balance of nature in an ecosystem through:
   ✓ Protection and preservation of natural resources.
   ✓ reducing the rate of pollutions

Answer: A

8. Monoculture affects the biodiversity by:
   A. Reducing number of vegetation species.
   B. Increasing the rate of species.
   C. Destructing wildlife in a forest.
   D. None of the above.

Explanation: **Monoculture** is the destruction of forest habitat and replacing it only by a single species. **Answer: A**

9. Conservation of natural resources refers to:
   A. Using natural resource on sustainable basis.
   B. Keeping natural resources as they are
   C. Conserving plant and animal only
   D. Conserving water and rain only

Explanation: Conservation refers to the wise use of natural resources on a sustainable basis. In other words, conservation is protection and preservation of natural environment.

Answer: A

10. Which of the following action would contribute towards conservation of both water and soil?
   A. Deforestation
   B. Desertification
   C. Reforestation
   D. Overgrazing

Explanation: Deforestation, desertification and overgrazing have negative impact on conservation. While, reforestation is the act of planting trees which has positive impact conservation of soil and water. **Answer: C**

11. Biodiversity of vegetation could be increased by:
   A. Transplanting plants from other countries.
   B. Producing more cultivated varieties.
   C. Conversion of grass land to crop lands.
   D. Recycling of used materials.

Explanation: Increasing cultivated variety of plants increase the biodiversity in vegetation. **Answer: B**

12. Which of the following strategies is most likely to allow the preservation of the highest biological diversity of Ethiopia?
   A. Keep rare animals in zoo.
   B. Protect sufficient areas with different ecological zones.
   C. Protect large area of the natural forest.
   D. All are possible. **Answer: D**

13. The present day definition of conservation of natural resources refers to:
   A. Keeping the natural resources as they are
   B. Using natural resources on a sustainable basis.
   C. Conserving animals and plants only.
   D. Conserving and watering only.

Explanation: Conservation of natural resources belong to the protection and preservation of natural resources on a sustainable basis.

Answer: B

14. Which of the following is a non-renewable natural resource?
    A. Forest
    B. Wild life
    C. Oil and mineral
    D. Water

Explanation: Forest, wild life and water and renewable natural resources. While, oil and mineral are non-renewable resources.

Answer: C

15. Natural resources are renewable if they can be produced indefinitely.
    A. True
    B. False

Explanation: True because renewable natural resources can be easily replaced and produced indefinitely.

Answer: A

16. Which of the following is NOT a component of biodiversity?
    A. Animals
    B. Microorganism
    C. Soil
    D. Vegetation

Explanation: Biodiversity is a set of living things including animals, vegetation and microorganisms.

Answer: C

17. Distinguish between:
    - ✓ Renewable and nonrenewable natural resources
    - ✓ Biodiversity conservation and mono culturing.

Explanation:

Renewable natural resources	**Non-renewable natural resources**
- are resources that exist and produced indefinitely - they can be renewed or replaced easily	- are resources that exist in a limited amount and they cannot be easily replaced
Biodiversity conservation	**Monoculturing**
- is preservation and protection of variety of species of living things	- replacing a forest habital or farm land having variety of species by a single type of species only

18. Define the following
    - ✓ Conservation
    - ✓ Extinction
    - ✓ Biodiversity
    - ✓ Monoculture

Explanation:

- **Conservation:** is the protection and preservation of natural resources on a sustainable basis.
- **Extinction**: is a complete elimination or death of a given species of organisms from the earth.
- **Biodiversity**: is the diversity of wide range of like forms with in a given habitat or ecosystem.
- **Monoculture**: is replacement of vegetation cover in an area only be a single species of vegetation.

19. Explain why **monoculturing** is **NOT** recommended in the conservation of natural resources.

Explanation:

Monoculturing is an act of destructing the variety of species (bio diversity) and replacing only by a single species of plant. Thus, it is against the biodiversity conservation and not environment friendly.

20. What do we call a measure of the wealth of species in a given place?

A. Biomass
B. Environment
C. Biodiversity
D. Ecosystem

Explanation: Biodiversity is a measure of wealth of species in a given area or ecosystem.

21. Which of the following statement is correct about the importance conservation in given ecosystem?

A. Conservation maintains pure air and water in ecosystem.
B. Conservation maintains healthy biodiversity in the ecosystem.
C. Conservation gives favour for single species in the ecosystem.
D. Conservation reduces the biodiversity from the ecosystem.

Explanation:

Conservation of natural resources maintains healthy biodiversity in the ecosystem.

Answer: B

Post Test I

I Choose the best answer from the given alternatives.

1. One of the following is ~~NOT~~ a renewable source of energy. Which one is it?
   A. Coal
   B. Water
   C. Petroleum
   D. Natural gas
2. One of the following is exotic to Ethiopia:
   A. Woira
   B. Wanza
   C. Bahirzaf
   D. Kosso
3. Which one of the following is the disadvantage of monoculture?
   A. Elimination of biodiversity.
   B. Support wide range of animals and plants
   C. Lack of beauty
   D. Risks of disease to spread rapidly.
4. The measure of wealth of species in an area is:
   A. Biosphere
   B. Ecology
   C. Biodiversity
   D. Community
5. The wise and proper use of natural resources to maintain its sustainability is:
   A. Deforestation
   B. Conservation
   C. Cultivation
   D. Afforestation
6. Which one of the following is NOT a way to conserve biodiversity?
   A. Reduction of pollution
   B. Reduction of loss of habitat
   C. Deforestation of vegetation
   D. Legal actions
7. Covering huge field with a single crop is known as
   A. Monoculture
   B. Living
   C. Nonliving
   D. always available
8. Conservation of natural resources:
   A. Protects our sanitation
   B. Protects our environment
   C. Reduces the number of species
   D. Reduces exotic plants and animals

9. Which of the following activities is considered as a method of conservation? Increasing:
   A. the size of national parks to protect wild life
   B. the spread of disease
   C. the breeding of wild life with domestic animals
   D. deforestation in the ecosystem
10. Which of the following statements is correct about renewable resources?
   A. Cannot be replaced or reused.
   B. Never lost no matter how much we use them
   C. Are mainly fossil fuels and minerals
   D. Are mainly living things and their products
11. Which of the following methods helps to conserve biodiversity?
   A. Replanting of lands with indigenous plants
   B. Burning of plants which have no immediate use
   C. Planting trees that take long time to produce seeds
   D. Covering more area of land with exotic plants
12. Renewable natural resources different from non – renewable natural resources because:
   A. they exist indefinitely
   B. they cannot be easily replaced
   C. they are nonliving things
   D. they are scare
13. Conservation of natural resources refers to:
   A. Excessive use of natural resources
   B. Preservation of natural resources
   C. Unwise use of natural resources
   D. Replacing a forest by single species

Part II: Give short answer for the following questions.

14. Identify the following as renewable or non – renewable natural resources.
   - Water
   - Coal
   - Animal
   - Air
   - petroleum
   - diamond
   - vegetation

15. Define:
   - Extinction
   - Mono culture
   - Biodiversity

11.3 Vegetation

Vegetation is the most determinant natural resource for an ecosystem. This can be due to several reasons. The vegetation cover varies from one climatic area to another. For example; desert biome is characterized by lack of having large number of trees; while, tropical rain forest areas are ever green and have variety of vegetations from small shrubs to big trees.

Our country Ethiopia has variety of vegetations in which most of them are domestic. While, there are also some exotic vegetation types in our country.

> ***Domestic plants***- are plants that grow in a given locality.
> ***Exotic plants***- are plants that are from other countries.

Importance of Plants

Plants are useful for human beings for and other living things for the following major reasons:

- Sources of food;
- Revenue from export products such as timber, coffee, flower, etc;
- Raw materials for building houses, schools, fences, etc;
- Sources of medicines. Several types of modern and traditional medicines are made from vegetations;

> In Ethiopia several types of vegetations are used as sources of medicines. Some of their examples include:
>
> - Koso for tape worm;
> - Yedega agazen for treating dysentery.

- Habitat for several animals;
- Making of furniture and household equipments;
- Stabilization of climate. Plants release oxygen in to the atmosphere and absorb carbondioxide from the atmosphere which plays vital role in stabilizing the climate in our surrounding.
- Recreational purposes. People recreate themselves by going to forests, parks and other places where large number of vegetations are found.

Human Effect on Vegetation

Vegetations have several importance for human beings and the rest of organisms. However, human impacts are largely affecting the vegetation cover of an area. The action of humans that reduces or illuminates the vegetation cover of an area is called **deforestation.**

Some of the ways through which human beings impact vegetation are:

- Cultivation of crops for food or commercial purposes.
- Producing and Selling timber;
- Overgrazing by domestic animals;
- Uncontrolled fire damage;
- Building of houses, house hold and office equipments;
- Release of pollutants in to the atmosphere. These include; sulfur dioxide (SO2) and nitrogen dioxide (NO2), etc.

Overgrazing – is a repeated heavy grazing over several years that result in deterioration of the plant to community.

Endemic Species of Plants in Ethiopia

The word endemic belongs to a species of an organism that is unique to a particular geographic location or a country.

Ethiopia has several species of endemic plants that are found nowhere all over the world. From around 7000 species of higher plants that are fund in Ethiopia, about 800 of them are endemic to Ethiopia.

Some examples of Endemic plant species in Ethiopia are:

- Teff (*Eragrostis teff)*
- Noug (*Guizota abysinica)*
- Enset (*Ensete ventricosum*)
- Ficus Vasta
- Zigba
- Juniper (Tid)
- Kerero
- Sembo
- Anchote,etc

The history of vegetation in Ethiopia

Ethiopia is a country where there is a natural blessing in which its vegetation cover is largely indigenous and endemic plants. However improper action of human being and natural catastrophes have largely affected and destructed the vegetation cover of our country.

The history of Ethiopian vegetation at different times can be summarized as:

- ***Emperor Zera Yakob (1434- 1468)***- organized collection of seed of indigenous plants suh as juniper (tid), olive (woira), and podo (zigba) from wof washa which he planted them in Menagesha mountain. This forest is preserved until now.
- ***Emperor Minelik II (1895)***- ordered introduction of Bahirzaf (Eucalyptus) to replace native vegetations. Currently Bahirzaf (Eucalyptus) tree is widely distributed and used in Ethiopia for various purposes especially for construction of houses. However, it is also found to damage the soil. Therefore, a huge emphasis should be given to replace exotic plants such as Bahirzaf by indigenous (local) plants.

Conserving the vegetation resources on a sustainable manner is important for our country.

Vegetation cover of Ethiopia at different times can be summarized as:

- Before---------- 35%
- 1952---------- 16%
- 1980---------- 3.6%
- 1987---------- 2.7%
- 1990---------- 2.4%

The above information shows that there has been a huge decline in Ethiopian vegetation cover overtime.

Conservation

Conservation is the wise use of natural resources on a sustainable basis. Various ways can be suggested to conserve the vegetation in Ethiopia. Towards this concerned bodies should participate actively. Some of those concerned bodies who are involved include:

- The government working with various bodies to conserve and replant endemic species;
- Research institutions like national herbarium in Addis Ababa University
- Individuals at different level.

There are individuals who have contributed a lot for the conservation of Ethiopian vegetation. Some include:

- Dr. Mesfin Tadesse (1997) suggested ways of conserving natural vegetations.

> ➢ Professor Legesse Negash worked on a means of replanting indigenous tree species.

Some of the ways of conserving the natural vegetation are:

1. Educating the society on ways of being careful about conserving vegetation resources. This includes fire prevention.
2. Afforestation and reforestation

> **Afforestation** – the process of planting trees on bare land.
> **Reforestation** – the process of planting native trees in to a forest that has decreasing numbers of trees.

3. Helping people those relied on vegetation resources for alternative ways of generating income.
4. Establishing conservation areas such as parks.

Illustrative Example

20. The biome with the highest number of vegetation species is:
    A. Grassland C. Tundra
    B. Tropical rain forest D. Savanna

    Explanation: Tropical rain forest is a biome with the highest amount of rain fall and biodiversity. **Answer: B**

21. Vegetation conservation can be considered to be a very fundamental aspect of the scientific practice of biological resource conservation because:
    A. It is economically the only important item of the environment.
    B. It is conservation results in the conservation of soil, water and wild life.
    C. It is a very important component of the ecosystem.
    D. It is destruction starts a chain of reactions leading to complete land degradation.
    E. All except A

    Explanation: Vegetation is economically important in various ways. However, it is not the only economically important resource in an environment. Natural resources such as wild life, cola, natural gas are also economically important resources of an environment. **Answer: E**

22. For which of the following environmental problems would mankind held least responsible?
    A. Deforestation of forest ecosystem

B. Overgrazing grassland ecosystem
C. Wind erosion in desert
D. Pollution in aquatic habitat

Explanation: Deforestation, overgrazing and pollution of aquatic habitat are environmental problems that are highly, linked to human activities. However, erosion by wind in desert habitat is more of natural phenomenon than human intervention.

Answer: C

23. Ethiopia has various types of vegetation. This is mainly due to its:
A. Tropical climate
B. Location and lakes
C. Nearness to equator
D. Varied altitude and rainfall

Explanation: Ethiopia is a country that is rich by its bio diversity including wildlife and vegetation. This can be related to several factors but mainly due to the reason that, it has varied altitude and rain fall.

Answer: D

24. Negative effect on vegetation include all except:
A. Afforestation
B. Deforestation
C. Overgrazing
D. Burning forest

Explanation:
Deforestation overgrazing and burning of forest have negative effect on vegetation. While, afforestation has positive relationship with vegetation conservation.

Answer: A

25. What is the main reason why vegetation is cut down by humans?
A. To provide open land for agricultural crops.
B. To provide space for agricultural crops.
C. For fire wood.
D. All are possible answers.

Explanation:
All are possible reasons why people destruct a forest habit.

Answer: D

26. How does monoculture affect biodiversity? It replaces:
A. A biodiversity rich area by a bare land.
B. One species by another species.
C. One species by many species.
D. A biodiversity rich area by a single crop plant.

Explanation: Monoculture is an act of replacing a biodiversity rich area by a single crop plant species.

Answer: D